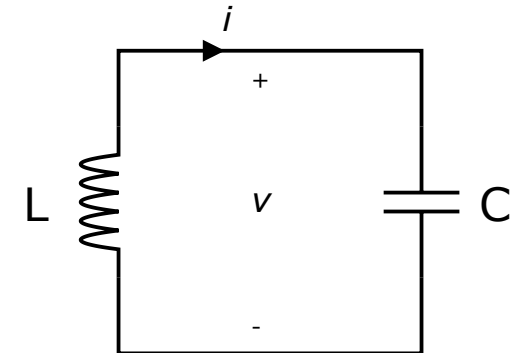
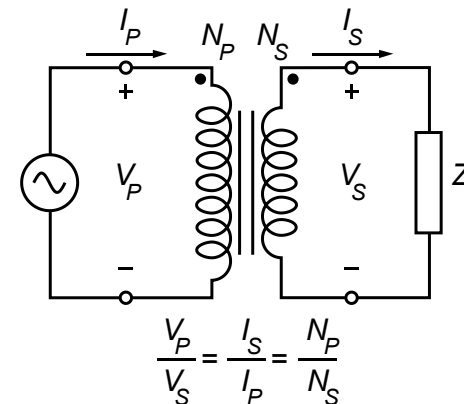
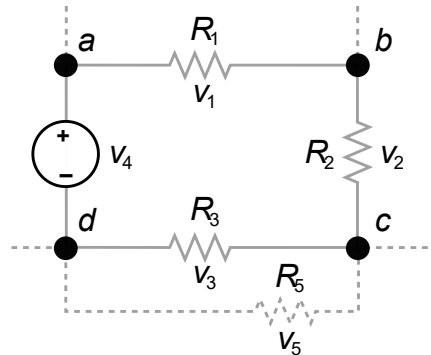
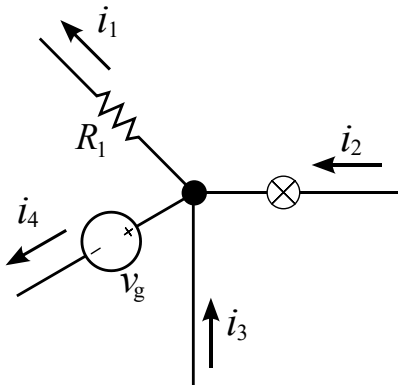


Introduction to Electrical Engineering



Overview

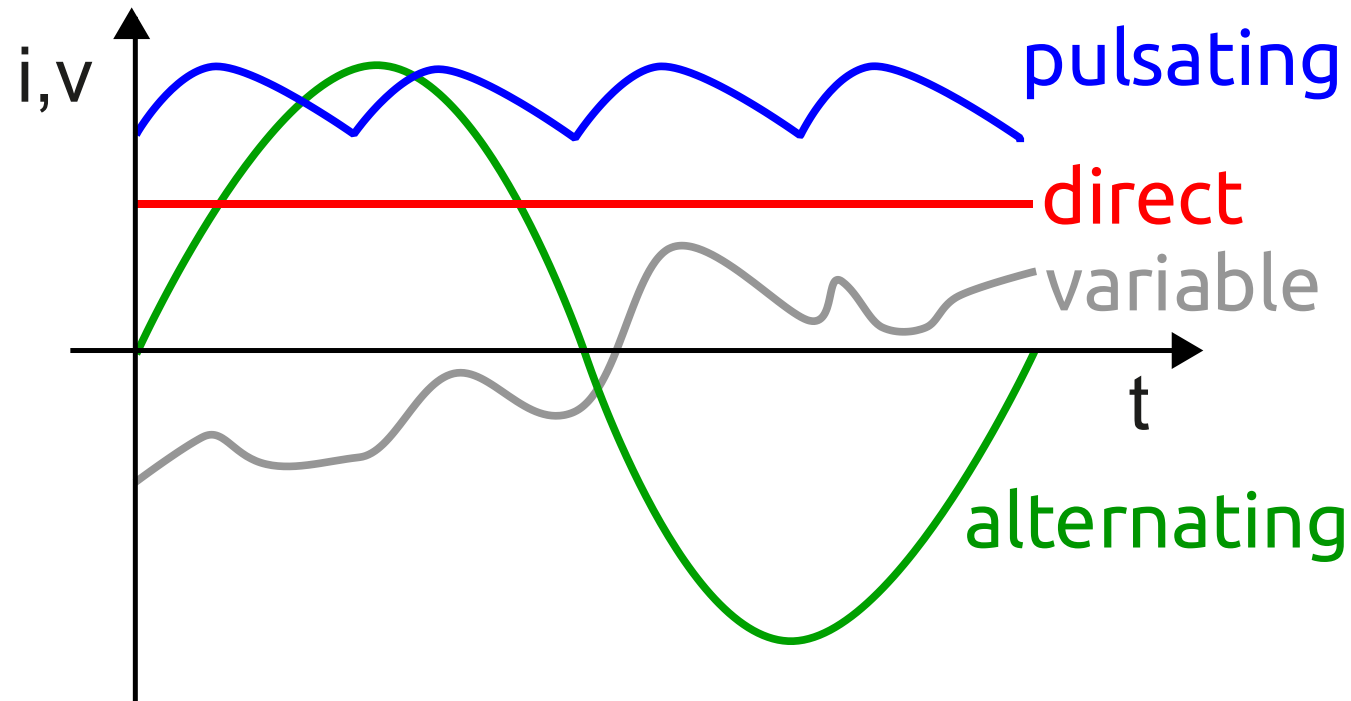
1. Physical Principles and Basic Concepts
2. Direct Current Circuits
3. Electric and Magnetic Fields
- 4. Alternating Current Technology**
5. Wrap-Up and Outlook

Alternating Current Technology

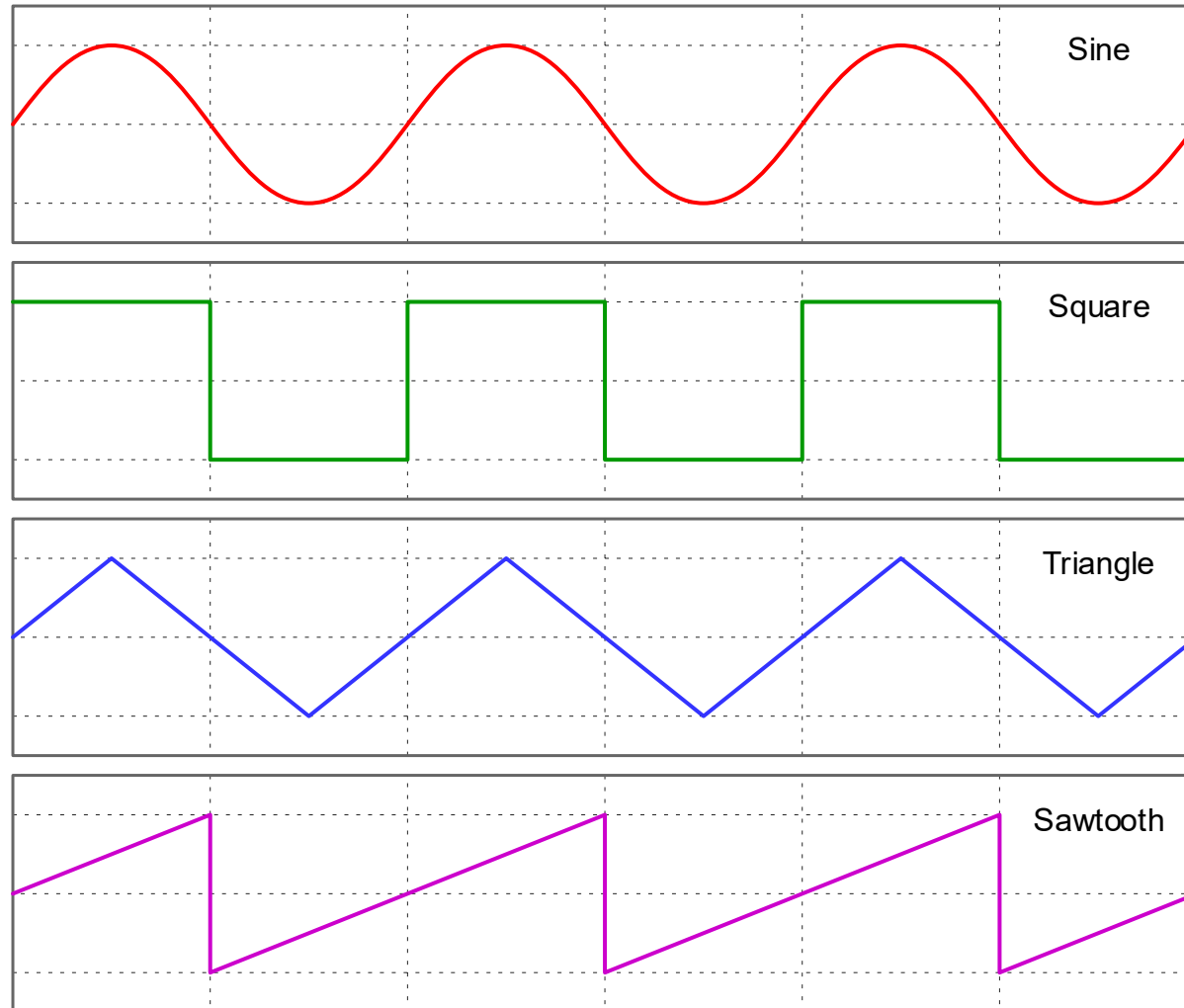
- Sinusoidal Waveform
- Phasor Representation
- Network Analysis in Phasor Domain
- AC Power
- High- & Low-Pass Filters
- Resonant Circuits
- Three-Phase Circuits

Alternating Current (AC)

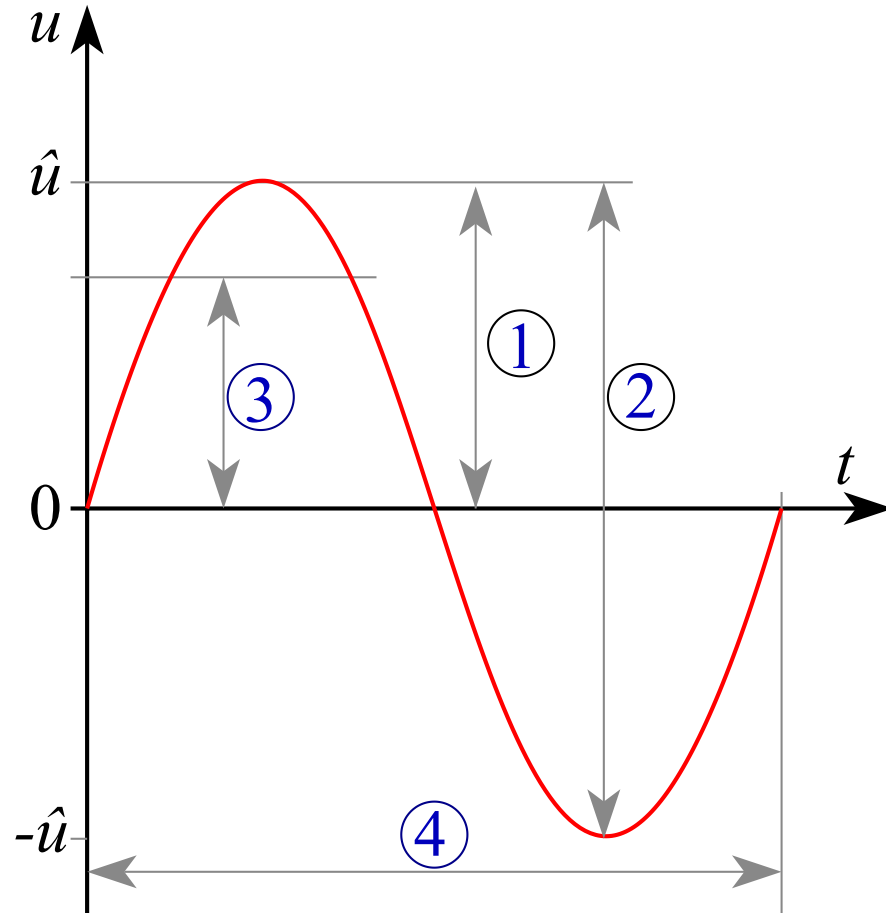
periodically reverses direction and changes its magnitude continuously with time



AC Waveforms



Sinusoidal AC



usual form of AC in most electric power circuits: sine wave

$$u(t) = \hat{u} \sin(\omega t), \quad i(t) = \hat{i} \sin(\omega t)$$

- $u(t) = u(t + nT)$
- frequency: $f = \frac{1}{T}$ $[f] = \frac{1}{s} = 1 \text{ Hz}$
- angular frequency: $\omega = 2\pi f$ $[\omega] = \frac{1}{s}$

1. peak voltage/current (amplitude) $\hat{u}, \hat{i}$
2. peak-to-peak amplitude: $2 \cdot \hat{u}$
3. RMS U, I
4. period (cycle time) T

Root Mean Square (RMS)

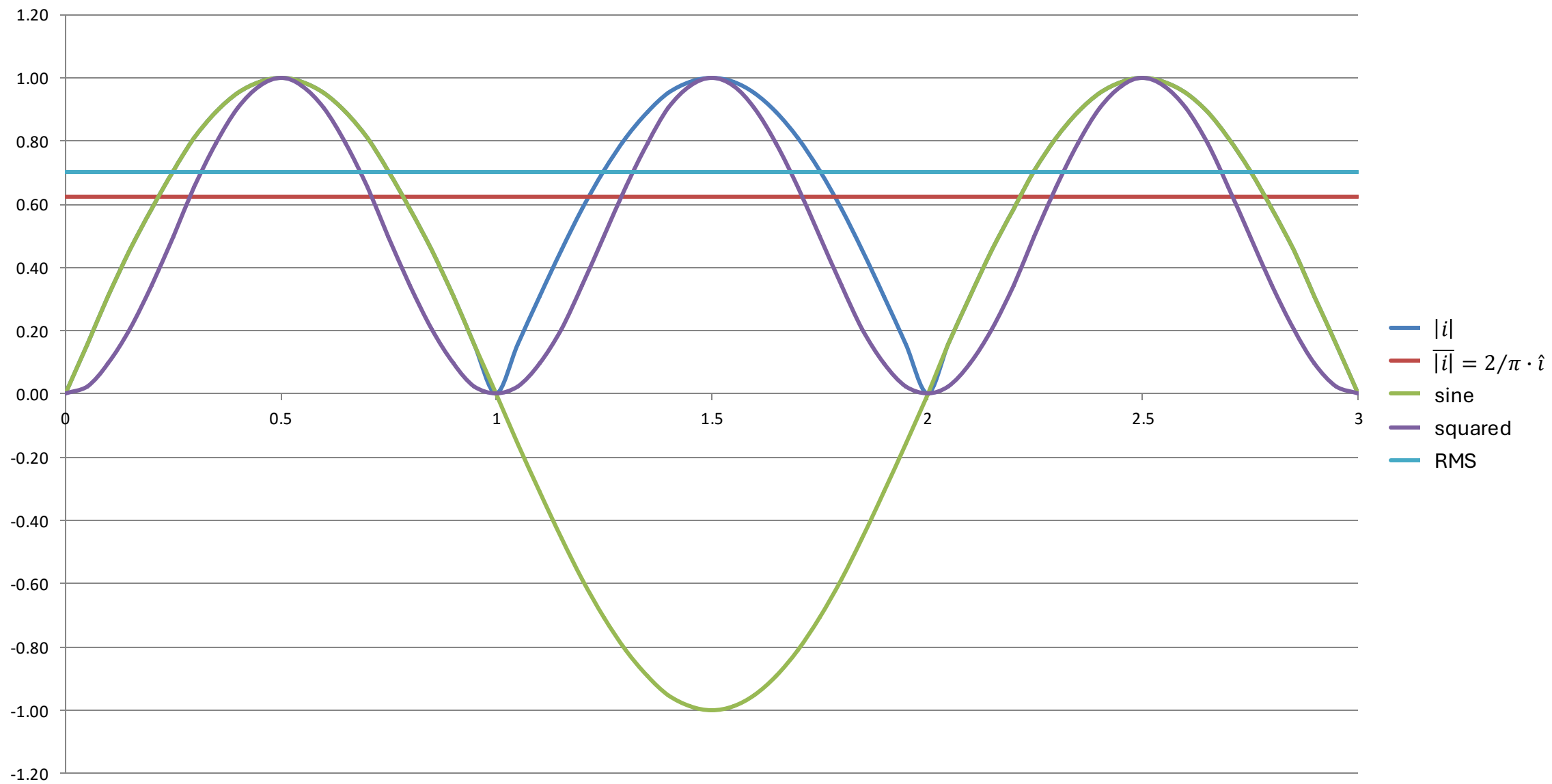
$$I = \sqrt{i^2} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} i^2(t) dt} = \frac{\hat{i}}{\sqrt{2}}$$

RMS of sum of mixed-frequency currents:
 $I_{\text{sum}} = \sqrt{I^2(f_1) + I^2(f_2)} \neq I(f_1) + I(f_2)$

The RMS of an AC corresponds to the DC value that generates the same amount of heat in a resistor. → effective value

mean itself is zero for pure AC:

$$\bar{i} = \frac{1}{T} \int_{t_0}^{t_0+T} i(t) dt = 0$$



Phase Shift

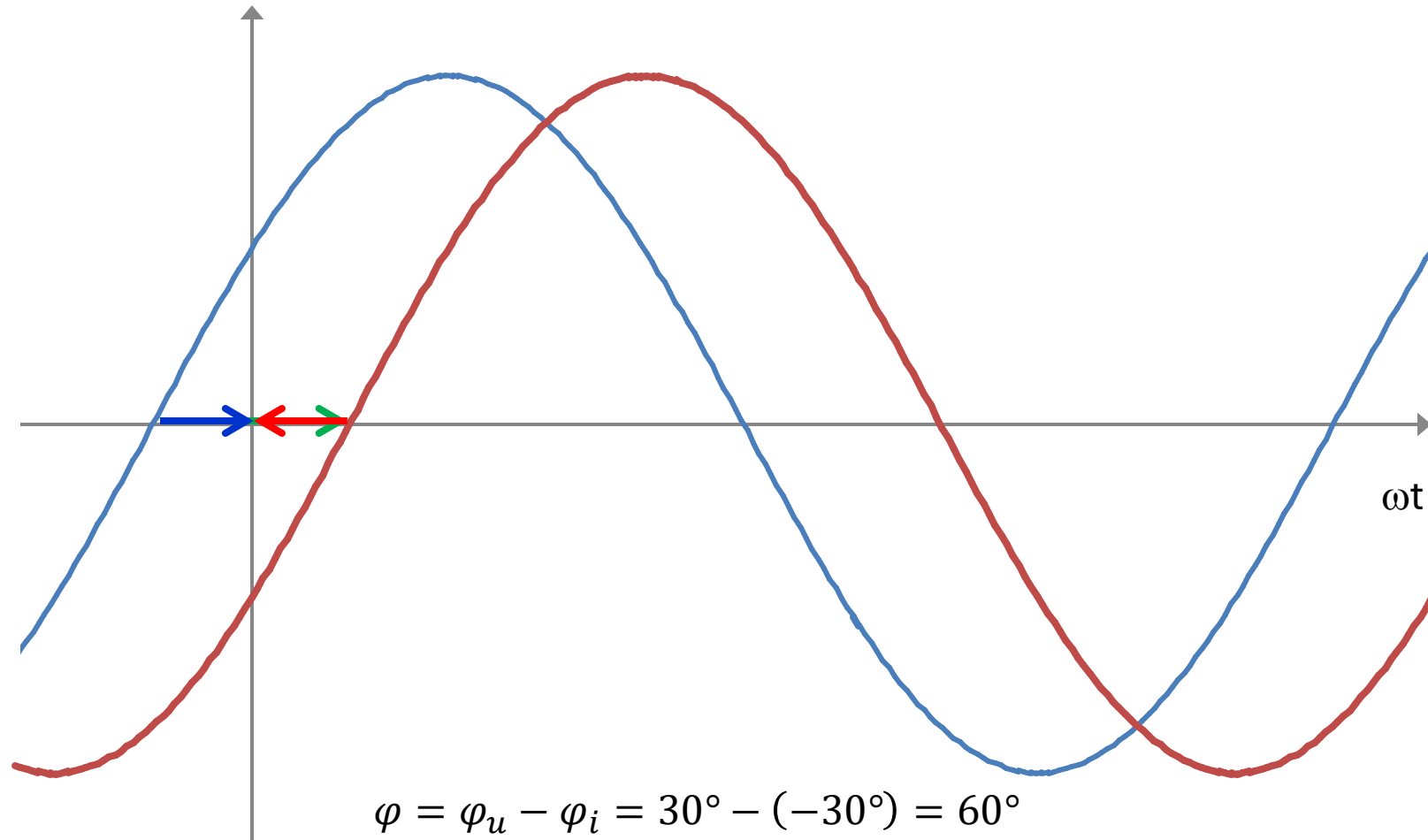
$$i(t) = \hat{i} \sin(\omega t + \varphi_0)$$

phase shift (for two currents with same frequency):

$$\varphi = \varphi_{01} - \varphi_{02}$$

phase shift between voltage and current (with current as reference):

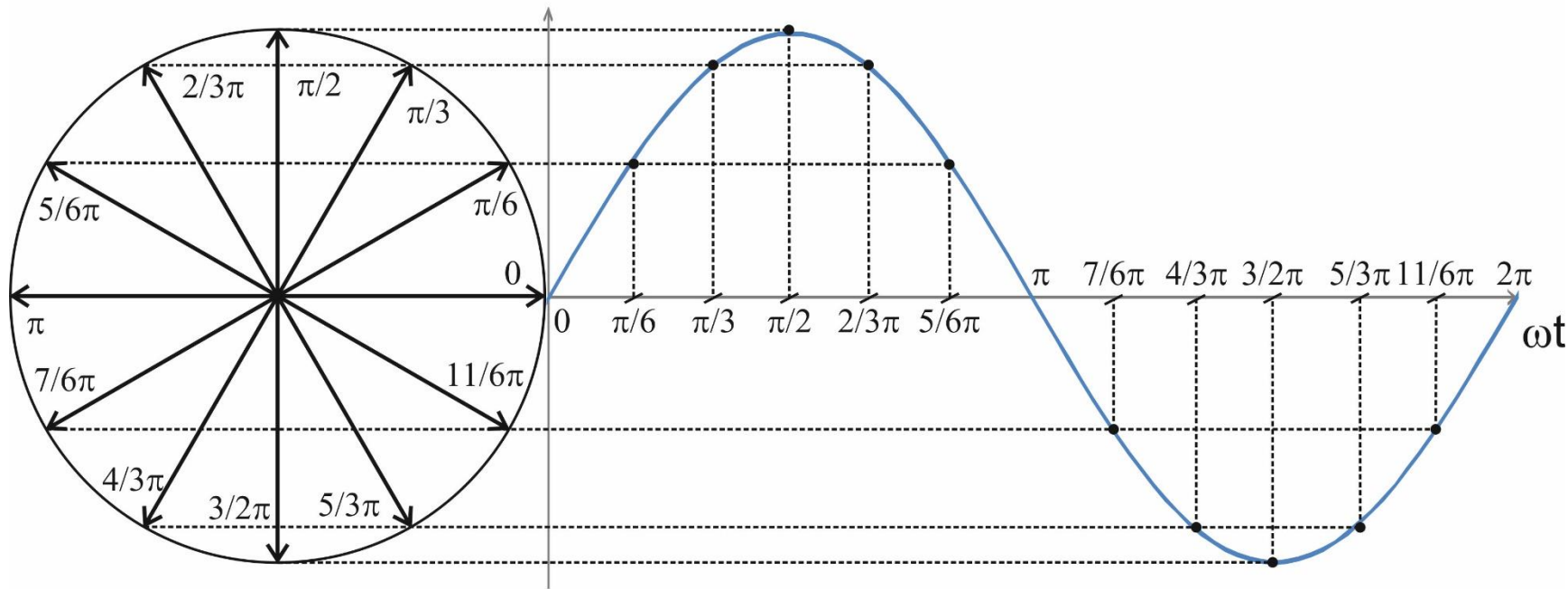
$$\varphi = \varphi_u - \varphi_i$$



In this example, the **current** lags behind the **voltage**.

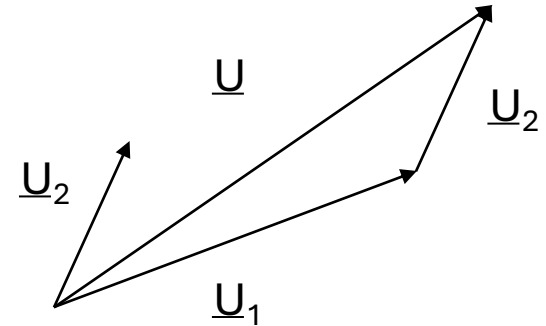
Phasor Representation

phasor diagram:

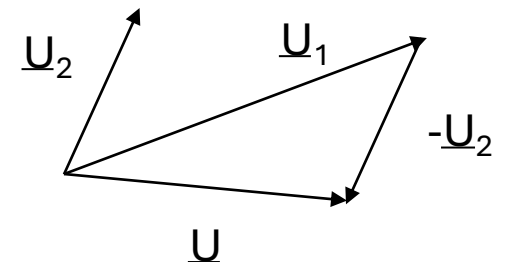


phasor calculations only work for quantities with same frequency
→ freeze rotation in time to get stationary phasors

addition:



subtraction:

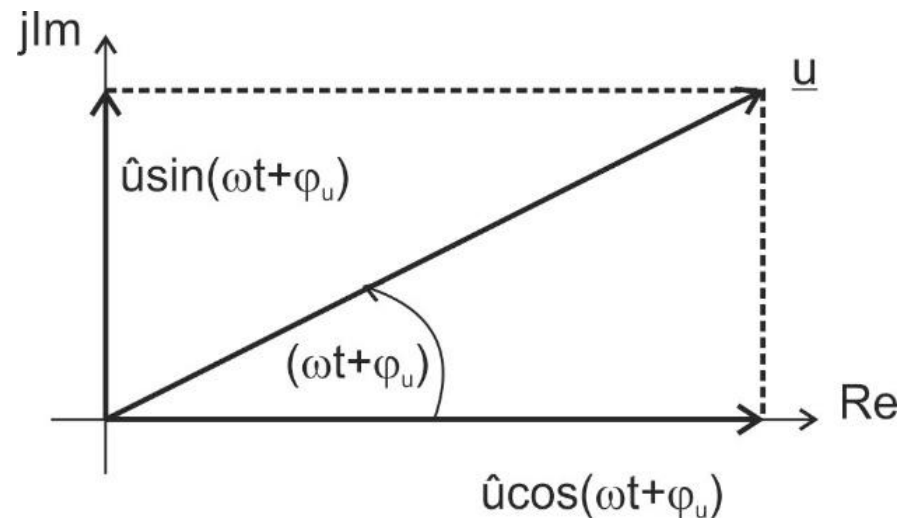


Use of Complex Numbers

$$u(t) = \hat{u} \cos(\omega t + \varphi_u) = \hat{u} \sin\left(\omega t + \varphi_u + \frac{\pi}{2}\right) = U \cdot \sqrt{2} \sin\left(\omega t + \varphi_u + \frac{\pi}{2}\right)$$

phasors can be represented by complex numbers:

$$\underline{u}(t) = \operatorname{Re}\{\underline{u}(t)\} + j \operatorname{Im}\{\underline{u}(t)\} = \hat{u} \cos(\omega t + \varphi_u) + j \hat{u} \sin(\omega t + \varphi_u)$$



Complex Amplitude

 $\underline{u}(t) = \hat{u}[\cos(\omega t + \varphi_u) + j \sin(\omega t + \varphi_u)] = \hat{u} e^{j(\omega t + \varphi_u)} = \hat{u} e^{j\varphi_u} \underbrace{e^{j\omega t}}$

complex

can be ignored if all quantities have the same frequency
→ stationary phasors

complex amplitude: $\underline{\hat{u}} = \hat{u} e^{j\varphi_u}$

$$\underline{u}(t) = \hat{u} e^{j\varphi_u} e^{j\omega t} = \underline{\hat{u}} e^{j\omega t}$$

complex RMS: $\underline{U} = U e^{j\varphi_u} = \frac{\underline{\hat{u}}}{\sqrt{2}}$

Transformation into Phasor Domain

time-dependent voltage	complex amplitude
$\hat{u} \cos(\omega t)$	$\underline{\hat{u}} = \hat{u}$
$\hat{u} \cos(\omega t + \varphi_u)$	$\underline{\hat{u}} = \hat{u} e^{j\varphi_u}$
$\hat{u} \sin(\omega t) = \hat{u} \cos\left(\omega t - \frac{\pi}{2}\right)$	$\underline{\hat{u}} = \hat{u} e^{-j\frac{\pi}{2}}$
$\hat{u} \sin(\omega t + \varphi_u) = \hat{u} \cos\left(\omega t + \varphi_u - \frac{\pi}{2}\right)$	$\underline{\hat{u}} = \hat{u} e^{j(\varphi_u - \frac{\pi}{2})}$

$$\underline{\hat{u}} = \hat{u}(\cos \varphi_u + j \sin \varphi_u) = Re + jIm$$

Back Transformation into Time Domain

1. multiplication with $e^{j\omega t}$ (phasors are stationary)
2. consider only real part of the complex number (or imaginary part if sine was used instead of cosine)

Derivation in Phasor Domain

$$\frac{d}{dt} \rightarrow j\omega$$

$$\left(\frac{d}{dt} e^{j\omega t} = e^{j\omega t} \cdot j\omega \right)$$

→ phase shift of $\frac{\pi}{2}$

$$\left(\underline{\hat{u}} \cdot j = \hat{u} e^{j\varphi_u} \cdot e^{j\frac{\pi}{2}} = \hat{u} e^{j(\varphi_u + \frac{\pi}{2})} \right)$$

example:

$$u(t) = L \frac{di}{dt} \rightarrow \underline{U} = j\omega L \underline{I}$$

Integration in Phasor Domain

$$\int dt \rightarrow \frac{1}{j\omega}$$

$$\left(\int e^{j\omega t} dt = e^{j\omega t} \cdot \frac{1}{j\omega} \right)$$

→ phase shift of $-\frac{\pi}{2}$

$$\left(\underline{\hat{u}}/j = \hat{u} e^{j\varphi_u} \cdot e^{-j\frac{\pi}{2}} = \hat{u} e^{j(\varphi_u - \frac{\pi}{2})} \right)$$

example:

$$i(t) = \frac{1}{L} \int u(t) dt \rightarrow \underline{I} = \frac{1}{j\omega L} \underline{U}$$

Complex Resistance

impedance:

$$\frac{\underline{u}(t)}{\underline{i}(t)} = \frac{\hat{\underline{u}} e^{j\omega t}}{\hat{\underline{i}} e^{j\omega t}} = \frac{|\hat{\underline{u}}| e^{j\varphi_u}}{|\hat{\underline{i}}| e^{j\varphi_i}} = |\underline{Z}| e^{j(\varphi_u - \varphi_i)} = |\underline{Z}| e^{j\varphi} = \underline{Z} \quad \text{abstract quantity}$$

Ohm's law in phasor domain: $\underline{\hat{u}} = \underline{Z} \underline{\hat{i}}$ or $\underline{U} = \underline{Z} \underline{I}$

$$\underline{Z} = R + jX = |\underline{Z}| e^{j\varphi}$$

resistance $\rightarrow R$

reactance $\rightarrow X$

impedance angle $\rightarrow \varphi$

modulus of impedance (apparent impedance) $\rightarrow |\underline{Z}|$

$$|\underline{Z}| = \sqrt{R^2 + X^2}$$

$$\tan \varphi = \frac{X}{R}$$

Complex Conductance

admittance:

abstract quantity

$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{1}{R + jX} = \underbrace{\frac{R}{R^2 + X^2}}_{\text{conductance } G} + j \underbrace{\frac{-X}{R^2 + X^2}}_{\text{susceptance } B}$$

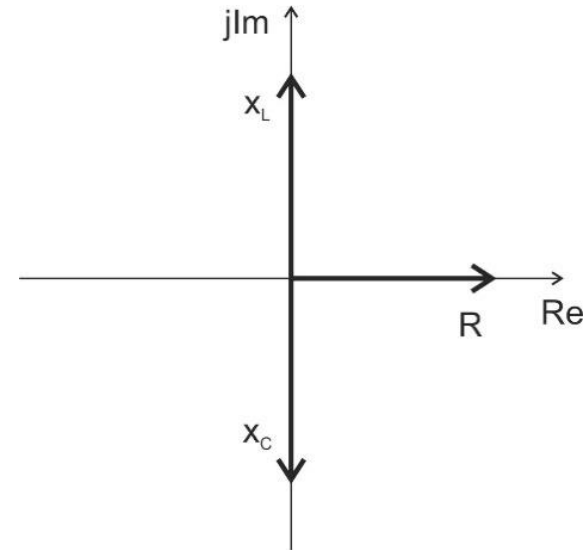
$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{1}{|\underline{Z}|e^{j\varphi}} = |\underline{Y}|e^{-j\varphi}$$

admittance angle
 $\theta = -\varphi$

modulus of admittance
(apparent admittance)

Linear Passive Components

R	$\underline{\hat{u}} = R \cdot \underline{\hat{i}}$	$\underline{Z} = R$
L	$\underline{\hat{u}} = j\omega L \cdot \underline{\hat{i}}$	$\underline{Z}_L = j\omega L = jX_L$
C	$\underline{\hat{u}} = \frac{1}{j\omega C} \cdot \underline{\hat{i}}$	$\underline{Z}_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C} = jX_C$

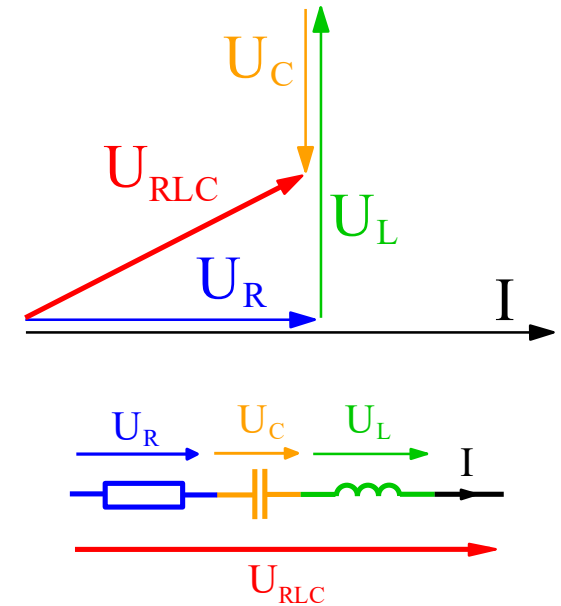


R: U and I in phase ($\varphi = 0$)

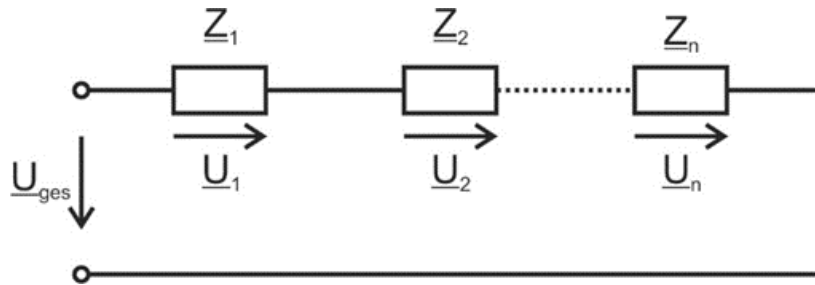
L: U leads I by 90°

C: U lags I by 90°

must consider angles:

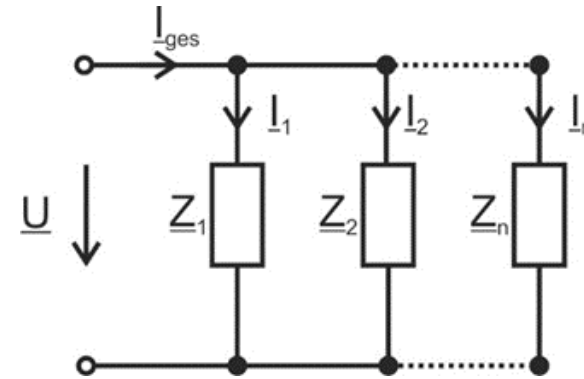


Network Analysis in Phasor Domain



$$\underline{Z} = \sum_{k=1}^n \underline{Z}_k$$

$$\frac{\underline{U}_1}{\underline{U}_2} = \frac{\underline{Z}_1}{\underline{Z}_2}$$



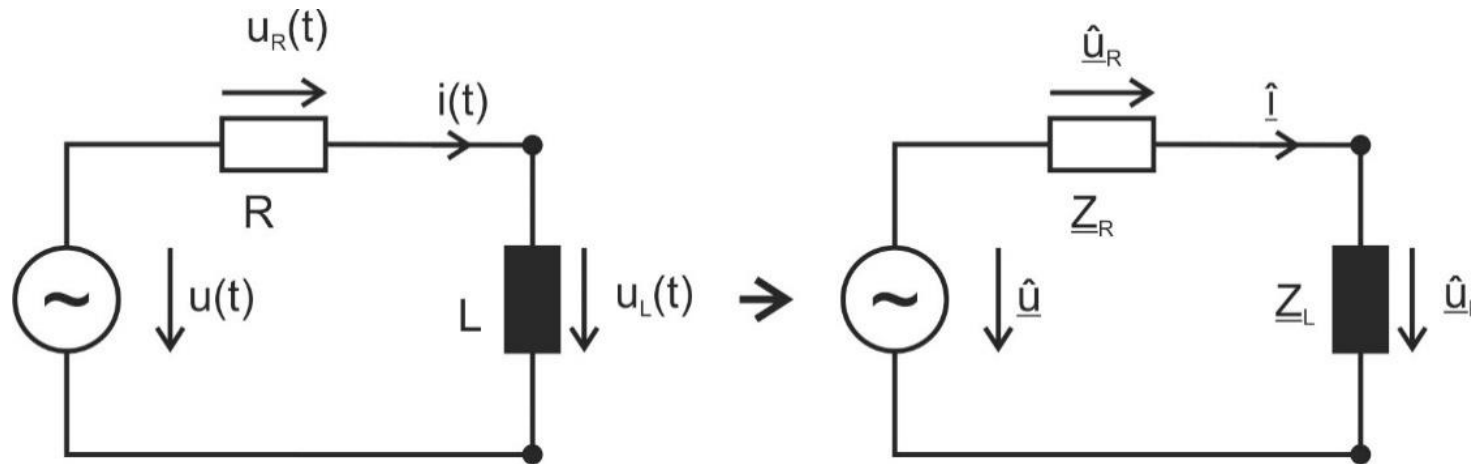
$$\frac{1}{\underline{Z}} = \sum_{k=1}^n \frac{1}{\underline{Z}_k}$$

$$\frac{\underline{I}_1}{\underline{I}_2} = \frac{\underline{Z}_2}{\underline{Z}_1}$$

KCL and KVL also valid for complex amplitudes and RMS values

Example 1

The current through the circuit and the voltage across the coil are to be determined.



$$u(t) = \hat{u} \cos(\omega t + \varphi_u) \rightarrow \underline{\hat{u}} = \hat{u} e^{j\varphi_u}$$

Example 1

$$\underline{\hat{u}} = \underline{\hat{u}}_R + \underline{\hat{u}}_L = \underline{Z}_R \cdot \underline{\hat{i}} + \underline{Z}_L \cdot \underline{\hat{i}} = (R + j\omega L)\underline{\hat{i}}$$

$$\underline{\hat{i}} = \frac{\underline{\hat{u}}}{R + j\omega L} = \frac{\underline{\hat{u}} \cdot (R - j\omega L)}{R^2 + \omega^2 L^2} = \underline{\hat{u}} \frac{R}{R^2 + \omega^2 L^2} - j\underline{\hat{u}} \frac{\omega L}{R^2 + \omega^2 L^2}$$

$$\hat{i} = \frac{\hat{u}}{Z} = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}}$$

$$\varphi = \tan^{-1} \frac{\omega L}{R} = \varphi_u - \varphi_i$$

$$\left(\varphi_u = \tan^{-1} \frac{\omega L}{R}; \quad \varphi_i = 0 \quad \text{or} \quad \varphi_u = 0; \quad \varphi_i = \tan^{-1} \frac{-\omega L}{R} \right)$$

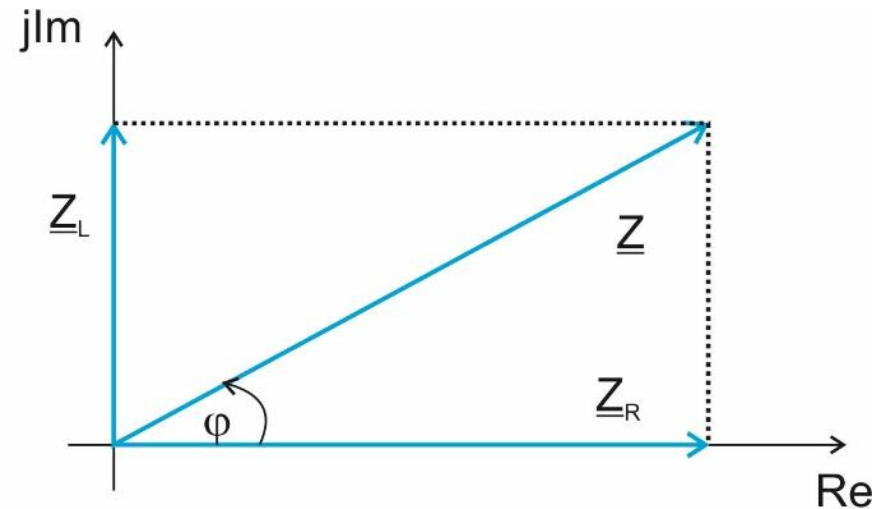
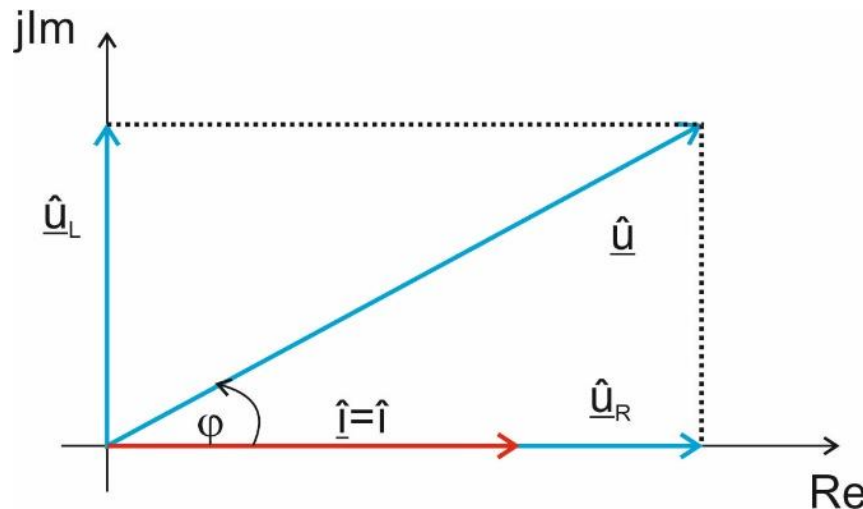
$$\underline{\hat{i}} = \hat{i} \cdot e^{j\varphi_i} = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}} e^{j(\varphi_u - \varphi)}$$

Example 1

$$i(t) = \text{Re}\{\underline{\hat{i}} \cdot e^{j\omega t}\} = \text{Re}\{\hat{i} \cdot e^{j\varphi_i} e^{j\omega t}\} = \hat{i} \cos(\omega t + \varphi_i) = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\omega t + \varphi_u - \tan^{-1} \frac{\omega L}{R}\right)$$

$$\underline{\hat{u}}_L = \underline{Z}_L \cdot \underline{\hat{i}} = j\omega L \underline{\hat{i}} = e^{j\frac{\pi}{2}} \omega L \hat{i} e^{j\varphi_i} = \hat{i} \omega L e^{j(\varphi_i + \frac{\pi}{2})}$$

$$u_L(t) = \text{Re}\{\underline{\hat{u}}_L e^{j\omega t}\} = \text{Re}\{\hat{i} \omega L e^{j(\omega t + \varphi_i + \frac{\pi}{2})}\} = \hat{u} \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\omega t + \frac{\pi}{2} + \varphi_u - \tan^{-1} \frac{\omega L}{R}\right)$$



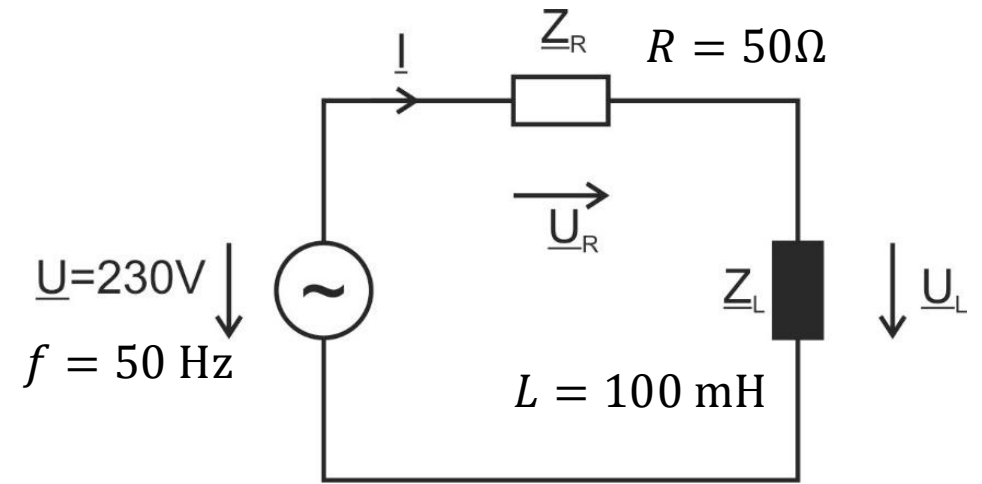
Example 1 with Numbers

$$\underline{Z} = \underline{Z}_R + \underline{Z}_L = R + j\omega L = (50 + j 31.42) \Omega$$
$$Z = \sqrt{R^2 + (\omega L)^2} = 59.1 \Omega$$
$$\varphi = \tan^{-1} \frac{\omega L}{R} = 32.1^\circ$$

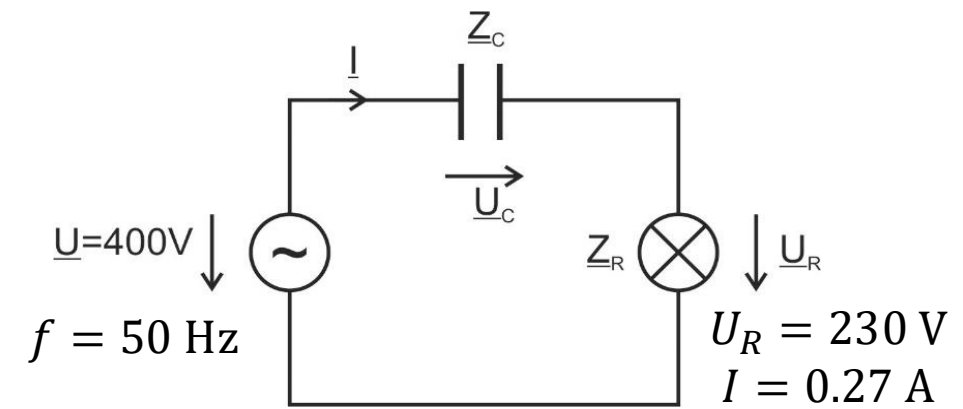
$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = \frac{U}{(R + j\omega L)} = \frac{U(R - j\omega L)}{(R^2 + (\omega L)^2)} = (3.30 - j 2.07) \text{ A} = 3.90 \text{ A} \cdot e^{-j 32.1^\circ}$$

$$\underline{U}_R = \underline{Z}_R \cdot \underline{I} = 50 \Omega \cdot 3.90 \text{ A} \cdot e^{-j 32.1^\circ} = 195 \text{ V} \cdot e^{-j 32.1^\circ} = (165.19 - j 103.63) \text{ V}$$

$$\underline{U}_L = \underline{Z}_L \cdot \underline{I} = j\omega L \cdot \underline{I} = j 31.42 \Omega \cdot (3.30 - j 2.07) \text{ A} = (65.04 + j 103.69) \text{ V} = 122.40 \text{ V} \cdot e^{j 57.9^\circ}$$



Example 2



What capacitance must the capacitor have so that the light bulb is connected to the voltage $U_R = 230\text{ V}$?

$$U^2 = U_C^2 + U_R^2 \rightarrow U_C = \sqrt{400^2 - 230^2}\text{ V} = 327.26\text{ V}$$

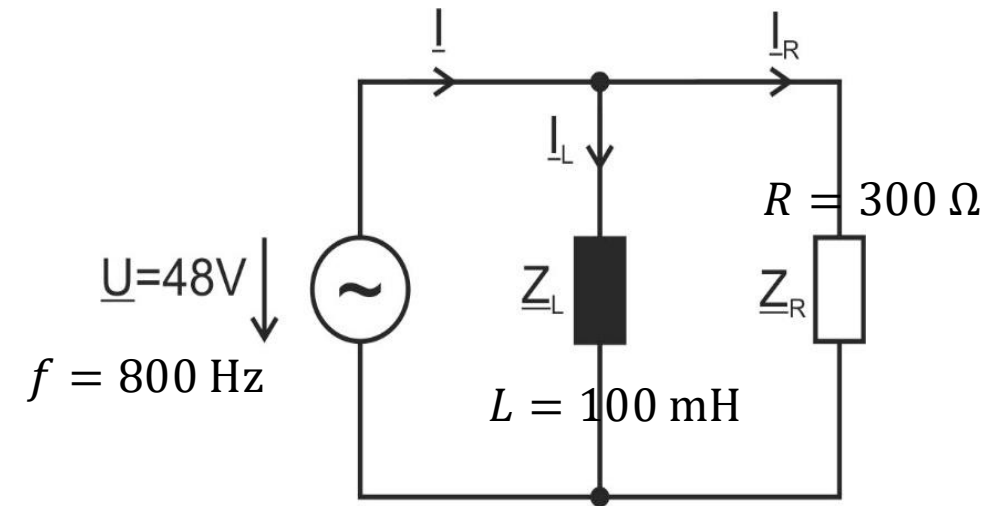
$$\frac{1}{\omega C} = \frac{U_C}{I} \rightarrow C = 2.63\text{ }\mu\text{F}$$

Example 3

$$\frac{1}{\underline{Z}} = \frac{1}{R} + \frac{1}{j\omega L} = \frac{j\omega L + R}{j\omega RL}$$

$$\rightarrow \underline{Z} = \frac{\omega^2 RL^2 + j\omega R^2 L}{R^2 + (\omega L)^2} = (221.2 + j 132.0) \Omega$$

$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = (160 - j 95) \text{ mA}$$



Instantaneous Power

$$p(t) = u(t) \cdot i(t)$$

For a purely ohmic resistance, current and voltage are in phase:

$$\begin{aligned} p(t) &= \hat{u} \sin \omega t \cdot \hat{i} \sin \omega t = \hat{u} \hat{i} \sin^2 \omega t \\ &= \hat{u} \hat{i} \frac{1}{2} (1 - \cos 2\omega t) = UI (1 - \cos 2\omega t) \end{aligned}$$

$p(t) > 0$ for passive sign convention $\rightarrow$ passive component

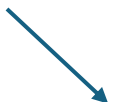
Average Power

area under power curve: electrical energy converted into heat in a resistor

average power (aka active power for resistance):

$$\bar{P} = \frac{1}{T} \int_0^T p(t) dt$$

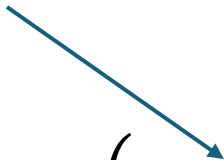
resistance


$$= \frac{1}{T} \int_0^T \frac{1}{2} \hat{u} \hat{i} (1 - \cos 2\omega t) dt = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(t - \frac{1}{2\omega} \sin 2\omega t \right) \Bigg|_0^T$$

$$= \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(T - \frac{1}{2\omega} \sin 2\omega T \right) = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(T - \frac{1}{2\omega} \sin 2\omega \frac{2\pi}{\omega} \right) = \frac{1}{2} \hat{u} \hat{i} = UI$$

Reactive Power

for reactance (capacitor or inductor): phase shift of $\frac{\pi}{2}$ between $u(t)$ and $i(t)$


$$p(t) = \hat{u} \sin \omega t \cdot \hat{i} \sin \left(\omega t + \frac{\pi}{2} \right) = \hat{u} \hat{i} \sin \omega t \cos \omega t = \hat{u} \hat{i} \frac{1}{2} \sin 2\omega t = UI \sin 2\omega t$$

$$\bar{P} = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{T} \int_0^T \frac{1}{2} \hat{u} \hat{i} (\sin 2\omega t) dt = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(-\frac{1}{2\omega} \cos 2\omega t \right) \Bigg|_0^T$$

$$= \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(-\frac{1}{2\omega} (\cos 2\omega T + 1) \right) = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(-\frac{1}{2\omega} (\cos 4\pi + 1) \right) = 0$$

→ not converted into heat or useful work

Apparent Power

mixed case for resistive-capacitive or resistive-inductive load:

$$p(t) = UI \cdot \cos \varphi \cdot (1 - \cos 2\omega t) - UI \cdot \sin \varphi \sin 2\omega t$$

- purely active for $\varphi = 0$
- purely reactive for $\varphi = \frac{\pi}{2}$ or $\varphi = -\frac{\pi}{2}$

average power:

$$P = UI \cdot \cos \varphi$$

apparent power:

$$S = U \cdot I$$
$$[S] = 1 \text{ VA}$$

electrical devices must be designed for the apparent power to be transmitted

Complex Power

$$\underline{S} = \underline{U} \cdot \underline{I}^* = U e^{j\varphi_u} I e^{-j\varphi_i} = S e^{j(\overbrace{\varphi_u - \varphi_i}^{\varphi})}$$

$$\underline{S} = \underline{U} \left(\frac{\underline{U}}{\underline{Z}} \right)^* = \underline{U} \cdot \underline{U}^* \frac{1}{\underline{Z}^*} = \frac{U^2}{\underline{Z}^*}$$

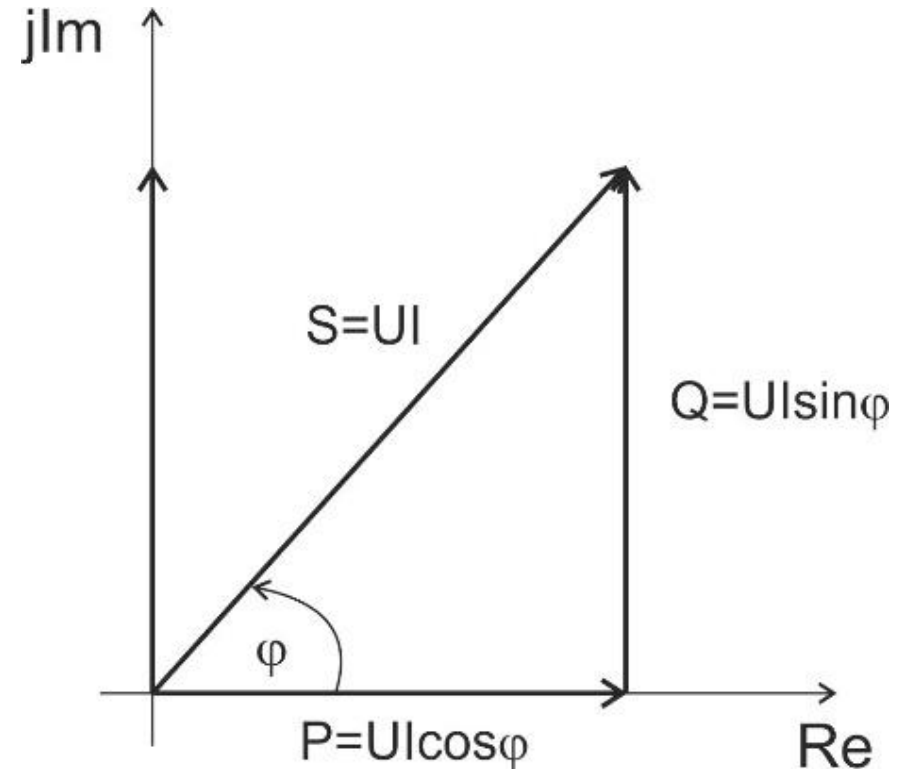
$$\underline{S} = \frac{1}{2} \hat{\underline{u}} \cdot \hat{\underline{i}}^* = P + jQ$$

active power

reactive power

with $\underline{S} = \sqrt{P^2 + Q^2} = U \cdot I$

apparent power

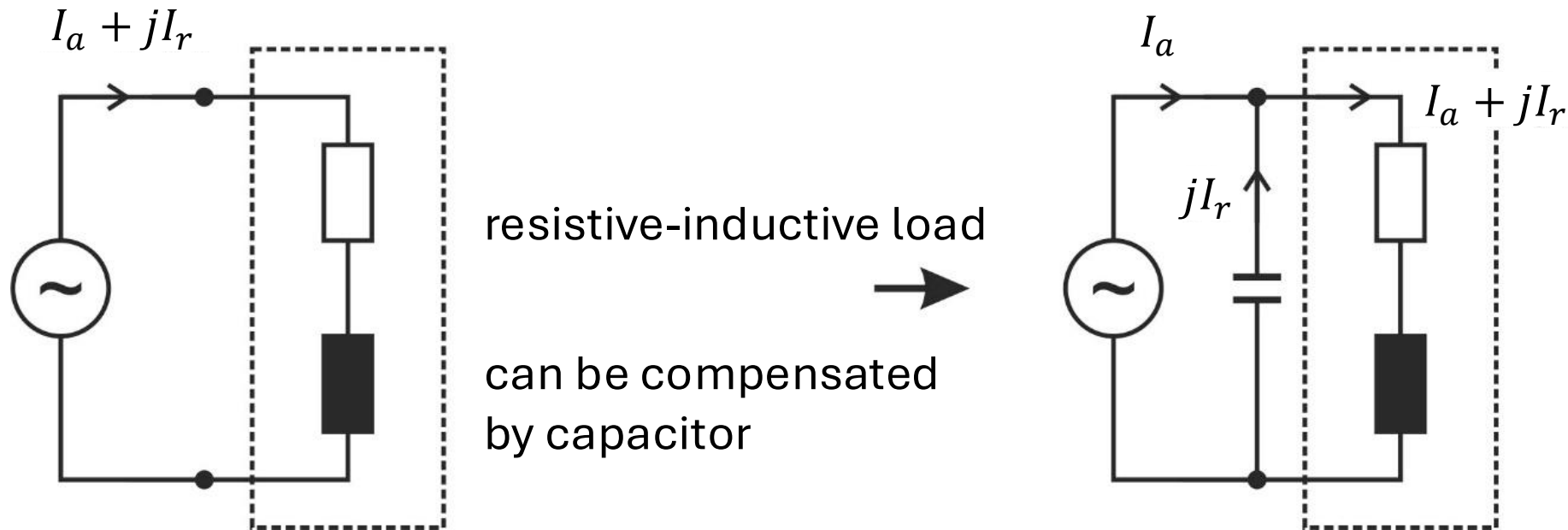


$$\varphi = \tan^{-1} \frac{Q}{P}; \quad \cos \varphi = \frac{P}{S}; \quad \sin \varphi = \frac{Q}{S}$$

Reactive Power Compensation

The absorption of inductive reactive power corresponds to the output of capacitive reactive power and vice versa: $\varphi = 180^\circ$

→ can compensate each other (useful to avoid transmission losses)



Example

A motor ($U = 230 \text{ V}$, $f = 50 \text{ Hz}$, $I = 16 \text{ A}$, $\cos \varphi = 0.8$) is to be compensated using a capacitor.

without capacitor:

$$I_a = I \cdot \cos \varphi = 12.8 \text{ A}, \quad I_r = I \cdot \sin \varphi = -9.6 \text{ A} \rightarrow \underline{I} = (12.8 - j 9.6) \text{ A}$$

$$\underline{S} = \underline{U} \cdot \underline{I}^* = 230 \text{ V} \cdot (12.8 + j 9.6) \text{ A}$$

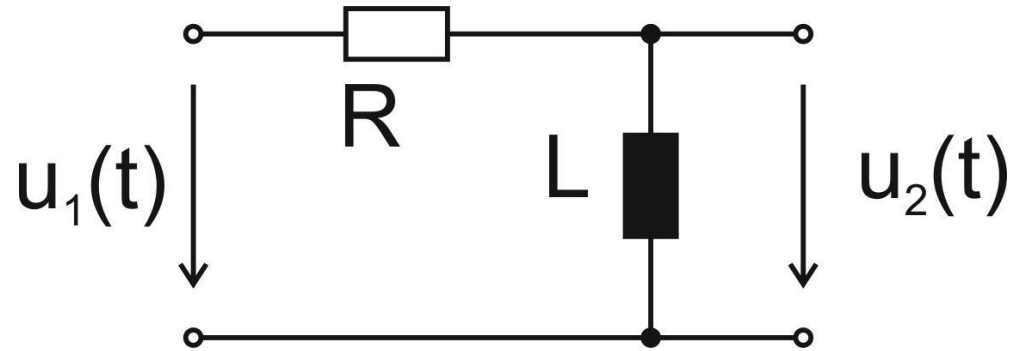
$$S = \sqrt{2.944^2 + 2.208^2} \text{ kVA} = 3.68 \text{ kVA}, \quad \varphi = \tan^{-1} \frac{2.208}{2.944} = 36.87^\circ$$

reactive current needs to be compensated by capacitor:

$$\frac{1}{\omega C} = \frac{U}{I_r} \rightarrow C = 133 \text{ } \mu\text{F}$$

$$\underline{S} = \underline{U} \cdot \underline{I}^* = 230 \text{ V} \cdot 12.8 \text{ A} = 2.944 \text{ kVA} = S, \quad \varphi = 0$$

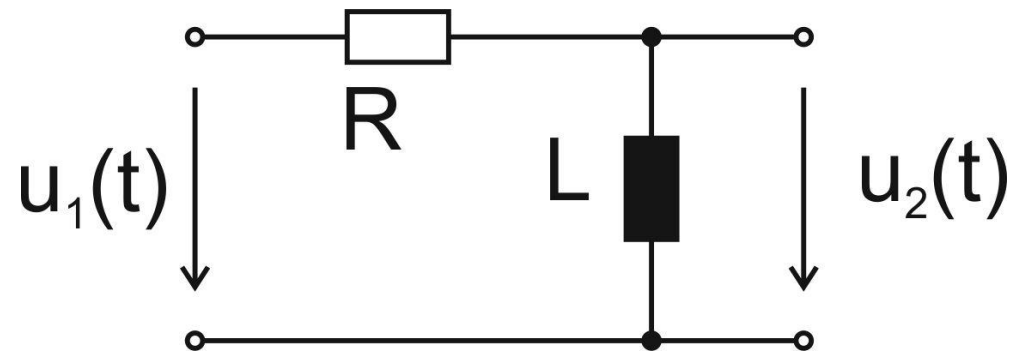
Frequency-Dependent Voltage Divider



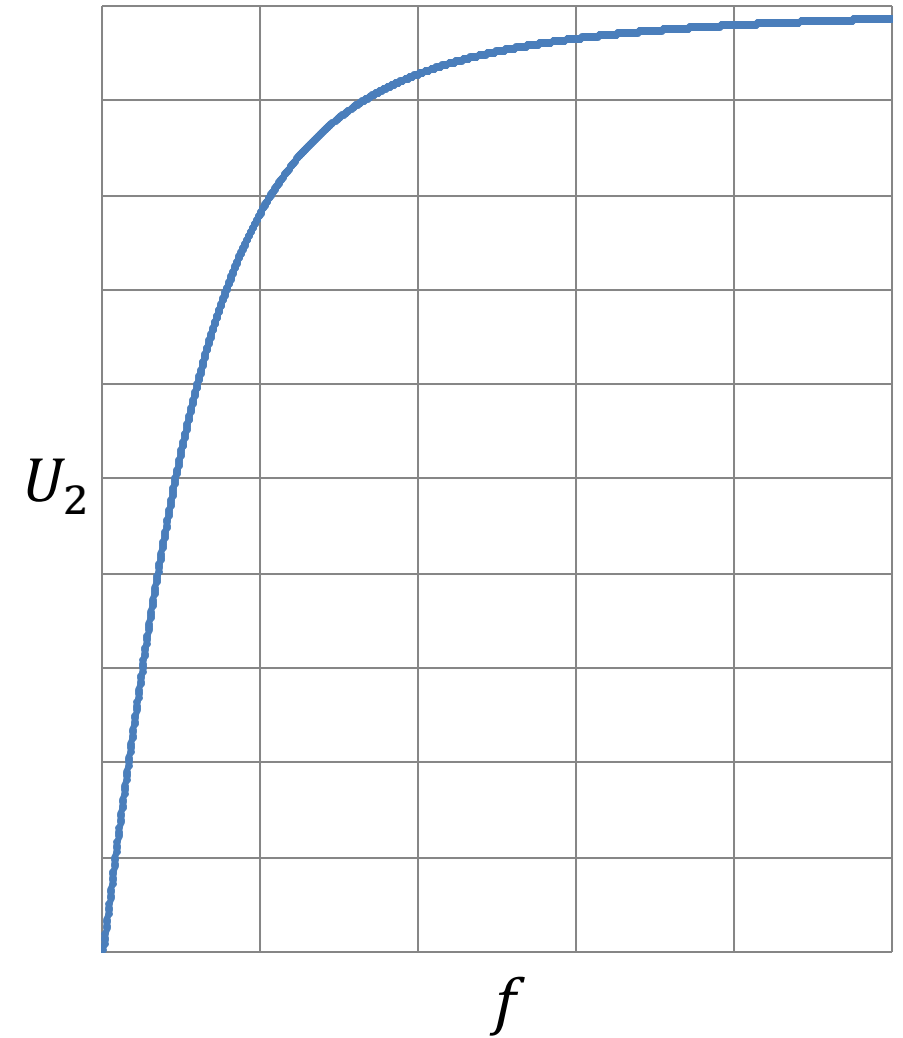
$$\underline{U}_2 = \underline{U}_1 \cdot \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} = \underline{U}_1 \cdot \frac{j\omega L}{R + j\omega L} = \underline{U}_1 \cdot \frac{(\omega L)^2 - j\omega LR}{R^2 + (\omega L)^2}$$

$$\begin{aligned} U_2 &= U_1 \cdot \sqrt{\left(\frac{(\omega L)^2}{R^2 + (\omega L)^2}\right)^2 + \left(\frac{\omega LR}{R^2 + (\omega L)^2}\right)^2} = U_1 \cdot \sqrt{\frac{(\omega L)^4 + (\omega LR)^2}{(R^2 + (\omega L)^2)^2}} \\ &= U_1 \frac{\sqrt{(\omega L)^2((\omega L)^2 + R^2)}}{R^2 + (\omega L)^2} = U_1 \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}} = U_1 \cdot \frac{Z_1}{Z} \end{aligned}$$

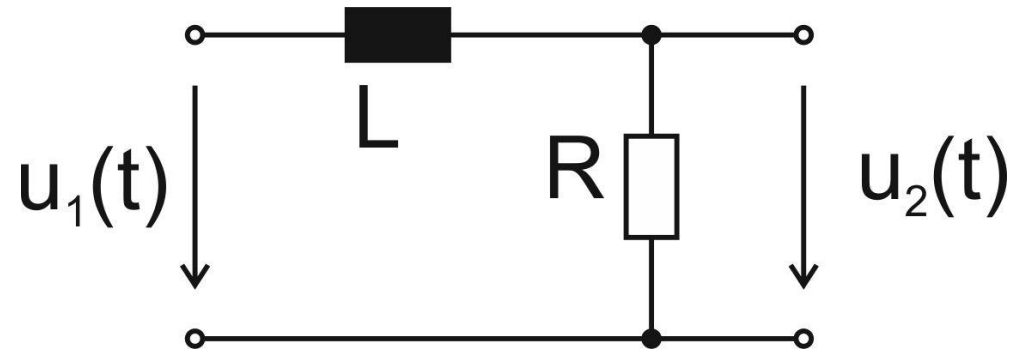
High-Pass RL Filter



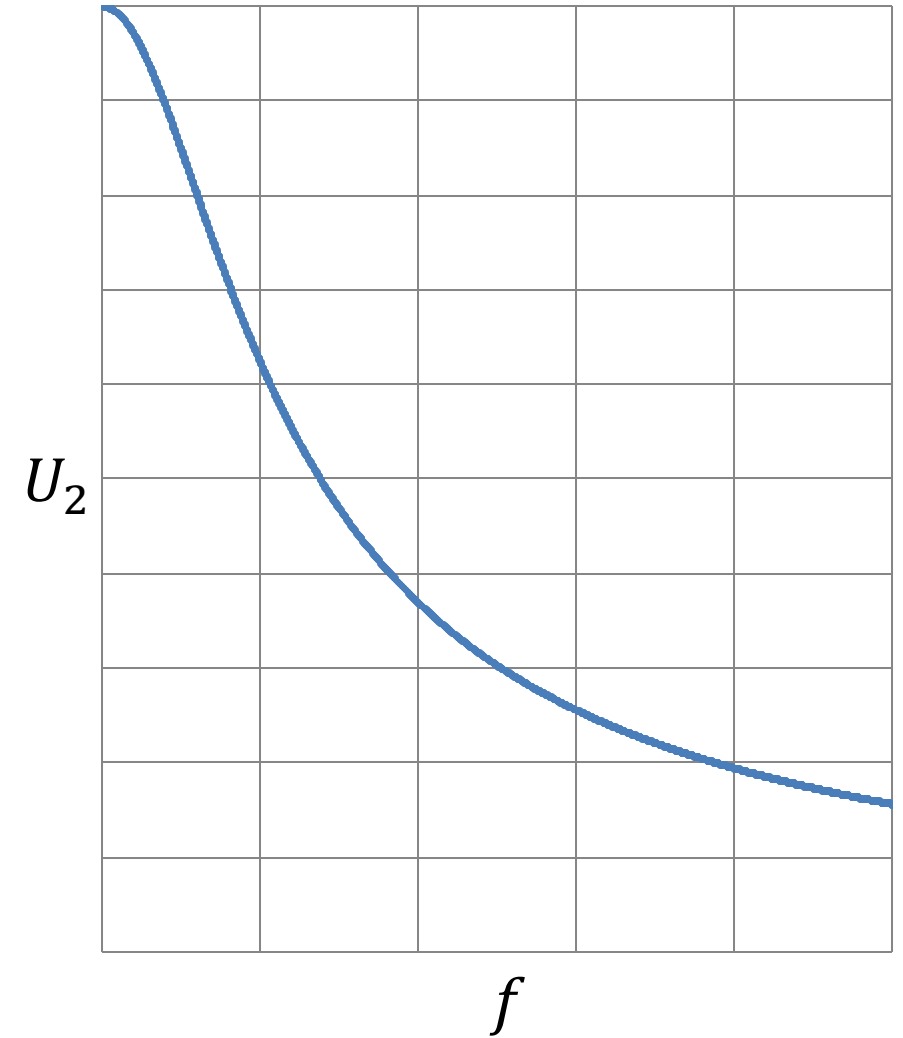
$$U_2 = U_1 \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}}$$



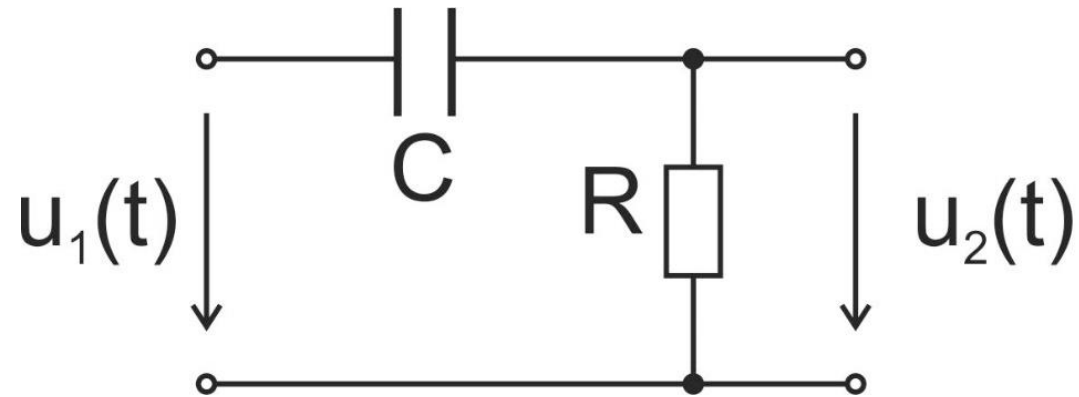
Low-Pass RL Filter



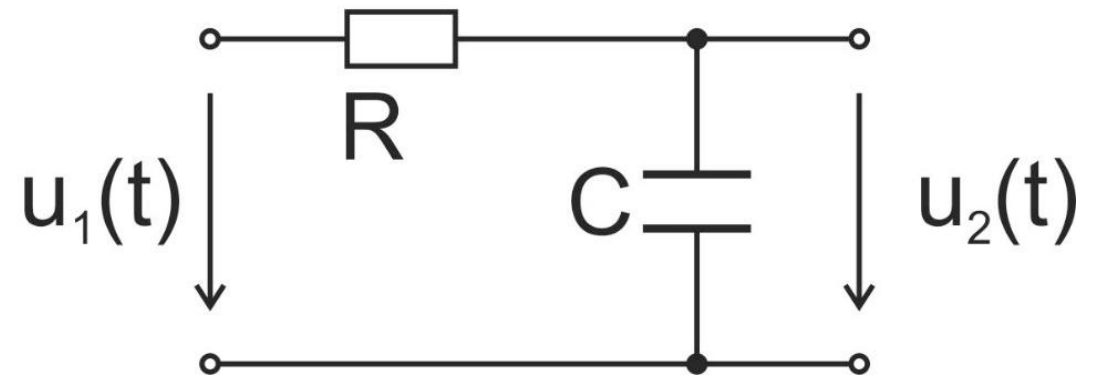
$$U_2 = U_1 \frac{R}{\sqrt{R^2 + (\omega L)^2}}$$



RC Filters



high-pass filter



low-pass filter

first-order filters: either a capacitor or an inductor

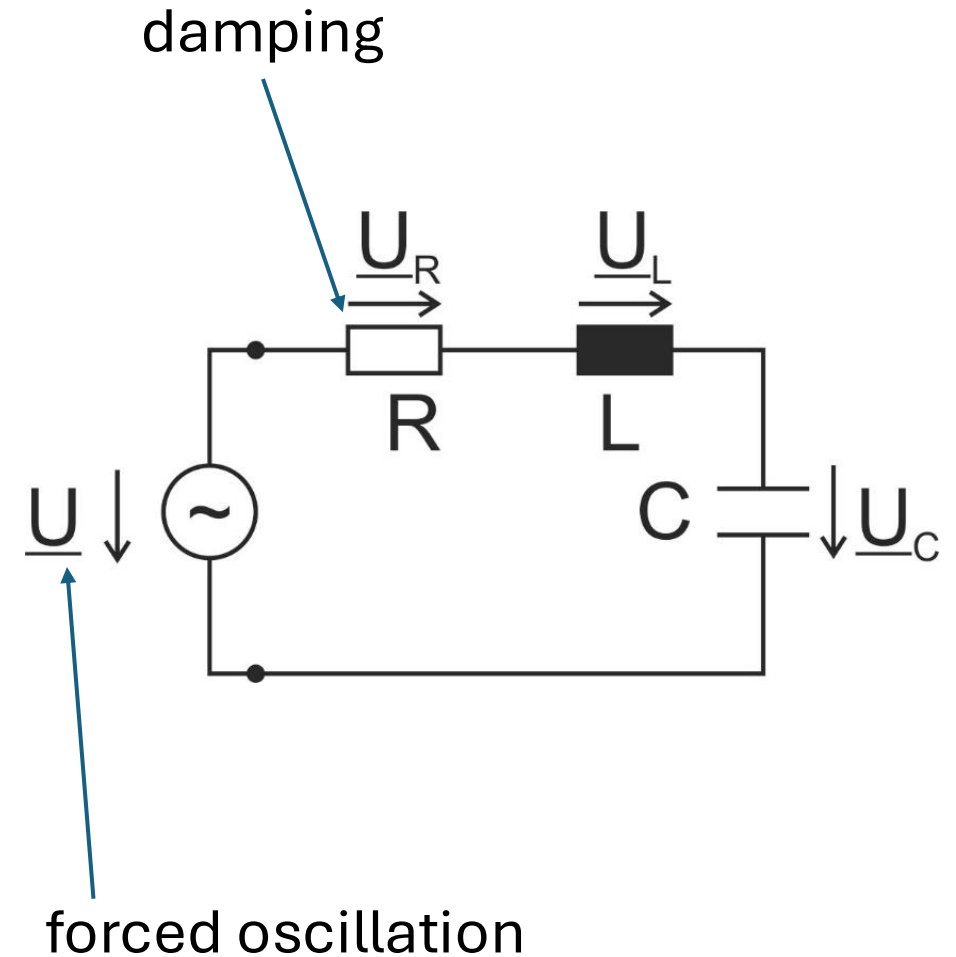
Series Resonant Circuit

$$\underline{Z} = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

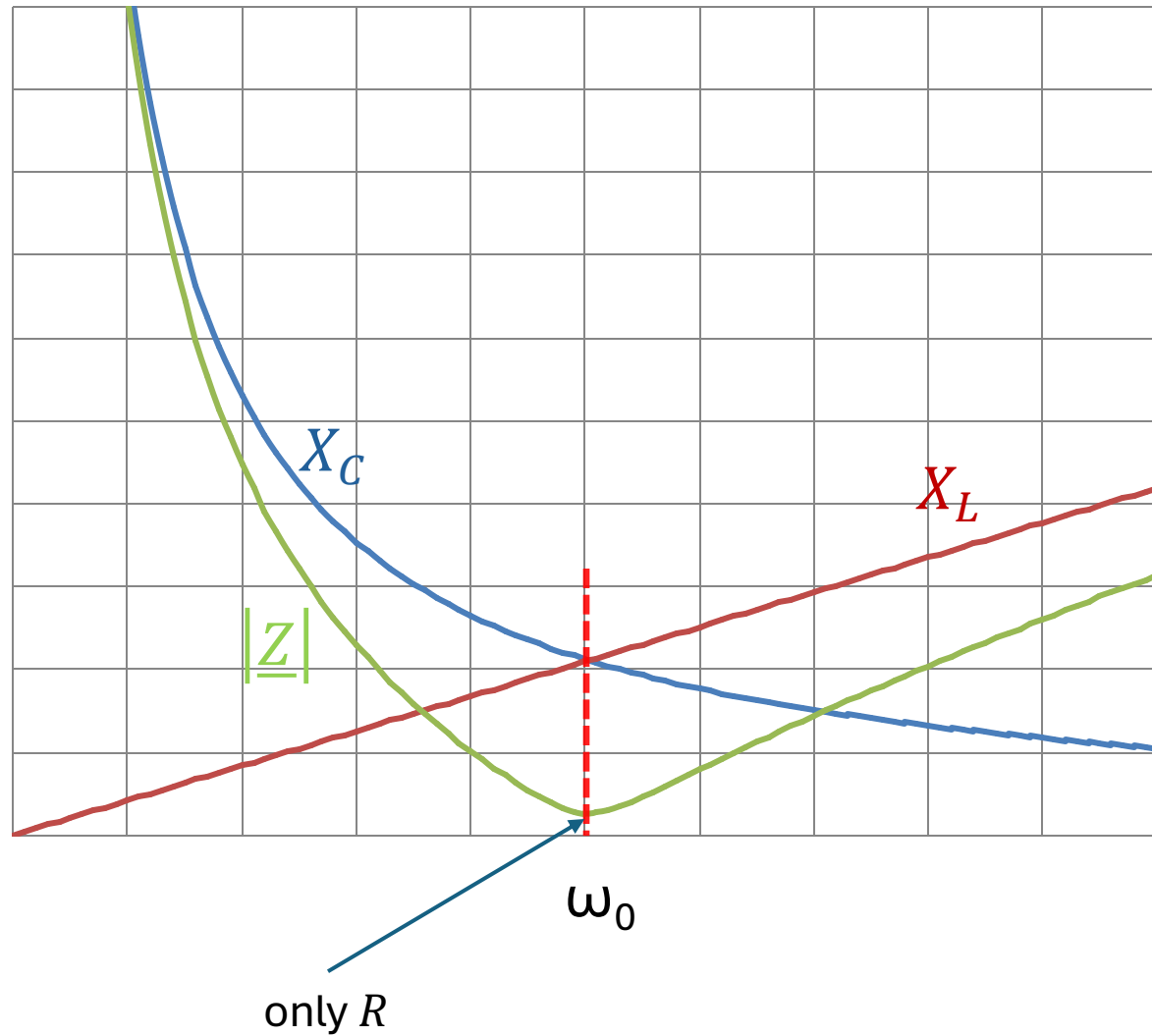
resonance frequency f_0 : inductive and capacitive reactances cancel each other out

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$
$$\rightarrow \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

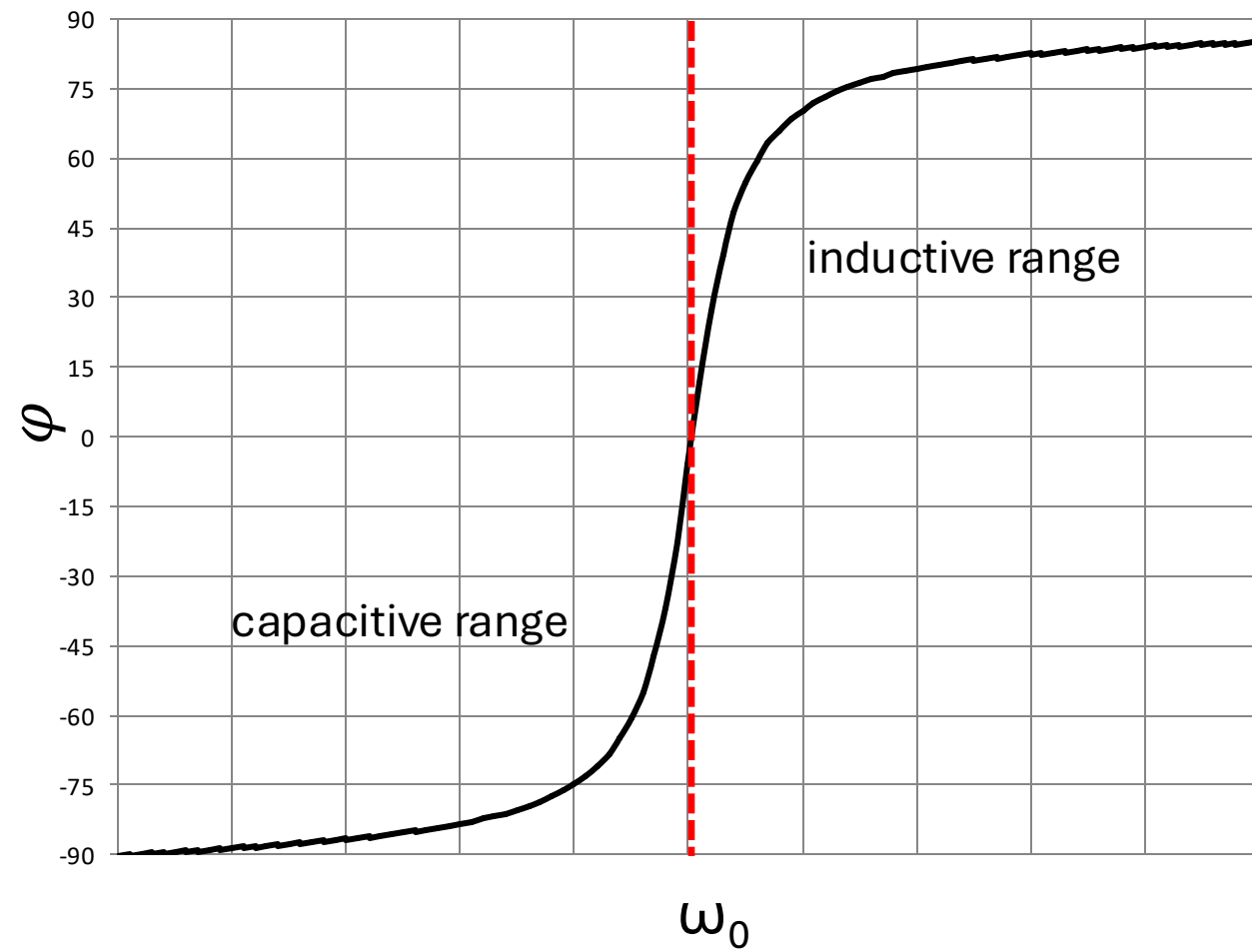
$$\underline{Z}(\omega_0) = R$$



impedance minimal at resonance:



phase shift between source voltage and current:



Voltage Resonance

$$U_L = U \frac{\omega L}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$U_C = U \frac{1}{\omega C \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

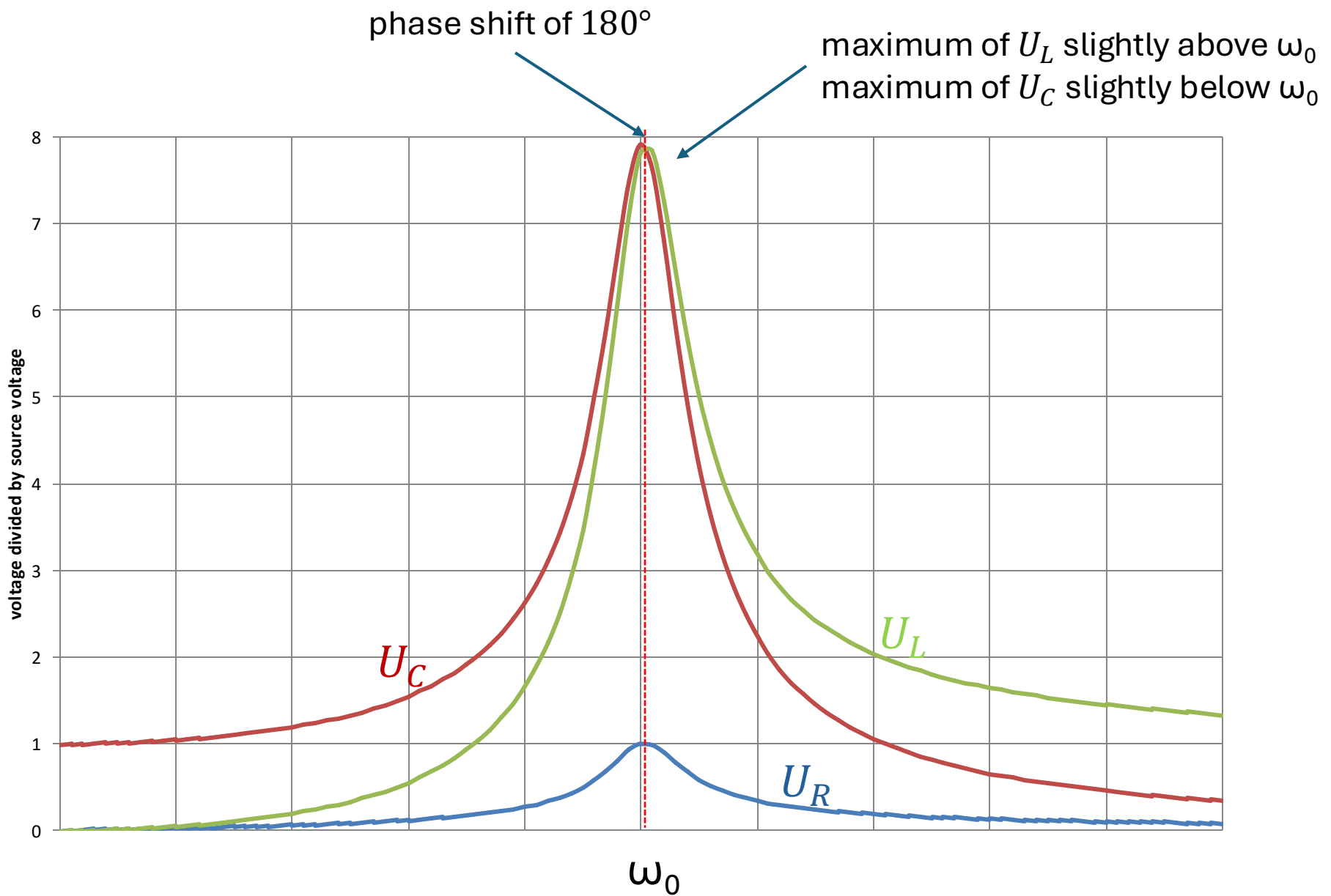
for ω_0 : $U_L = U_C = U \frac{1}{R} \sqrt{\frac{L}{C}} = U \cdot Q$

$$\underline{U}_L = -\underline{U}_C$$

quality factor $Q = \frac{1}{R} \sqrt{\frac{L}{C}}$

can be > 1

→ voltages across inductor and capacitor can be significantly larger than the source voltage (resonance catastrophe)



$$U_R = U \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \rightarrow \text{maximum for } \omega = \omega_0: U_R = U \quad (\text{also maximum current})$$

Voltage Maxima

$$\frac{dU_L}{d\omega_L} = 0 \quad \rightarrow \quad \omega_L = \sqrt{\frac{2}{2LC - R^2 C^2}} = \sqrt{\frac{2}{2LC - \frac{LC}{Q^2}}} = \omega_0 \frac{1}{\sqrt{1 - \frac{1}{2Q^2}}}$$

$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$ $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\frac{dU_C}{d\omega_C} = 0 \quad \rightarrow \quad \omega_C = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}} = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$$

→ no maxima for $Q \leq \frac{1}{\sqrt{2}}$

resonance behavior more pronounced the higher Q

Parallel Resonant Circuit

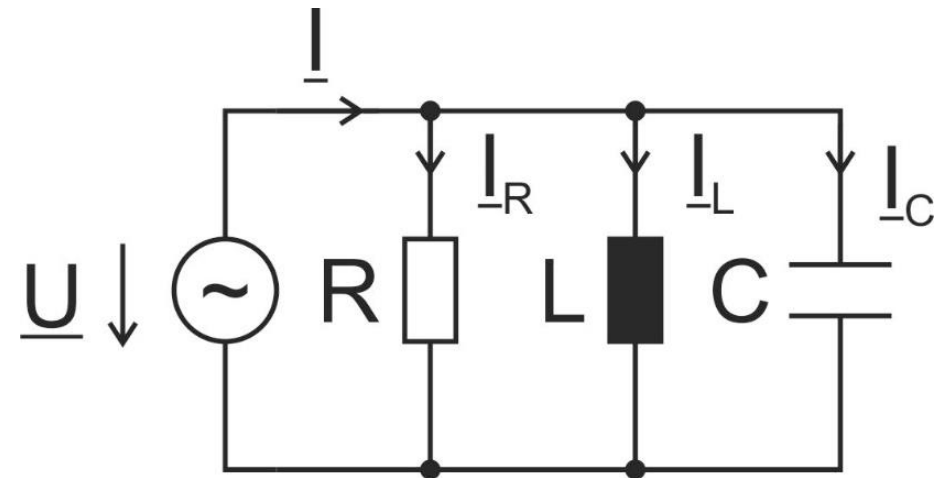
$$\underline{Y} = G + j \left(\omega C - \frac{1}{\omega L} \right)$$

$$\omega_0 C - \frac{1}{\omega_0 L} = 0$$
$$\rightarrow \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\underline{Y}(\omega_0) = G$$

admittance minimal at resonance (impedance maximal) $\rightarrow$ total current minimal ($\underline{I}_L = -\underline{I}_C$)

below ω_0 : determined by inductance
above ω_0 : determined by capacitance



Current Resonance

$$I_R = I \frac{G}{\sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

$$I_L = I \frac{1}{\omega L \sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

$$I_C = I \frac{\omega C}{\sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

resonance:

$$I_R = I = \frac{U}{R}$$

$$I_L = I_C = IR \sqrt{\frac{C}{L}} = I \cdot Q$$

→ current through inductor and capacitor can be larger than the source current

Current Maxima

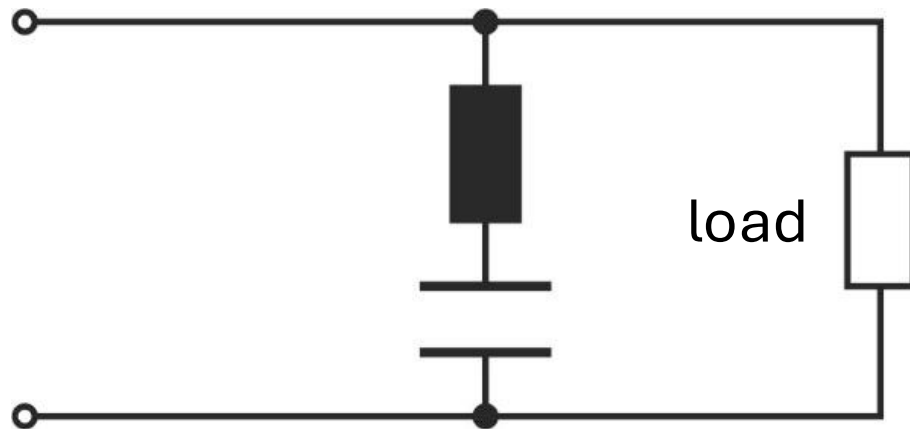
$$\frac{dI_L}{d\omega_L} = 0 \quad \rightarrow \quad \omega_L = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$$

$$\frac{dI_C}{d\omega_C} = 0 \quad \rightarrow \quad \omega_C = \omega_0 \frac{1}{\sqrt{1 - \frac{1}{2Q^2}}}$$

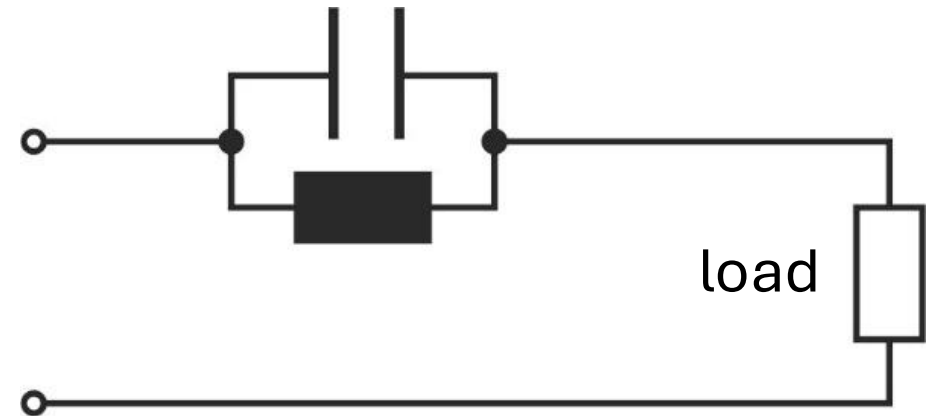
$\rightarrow$ again, no maxima for $Q \leq \frac{1}{\sqrt{2}}$

Second-Order Filters

acceptor circuit:
admittance peak at resonance



rejector circuit:
impedance peak at resonance



Three-Phase Circuits

idea:

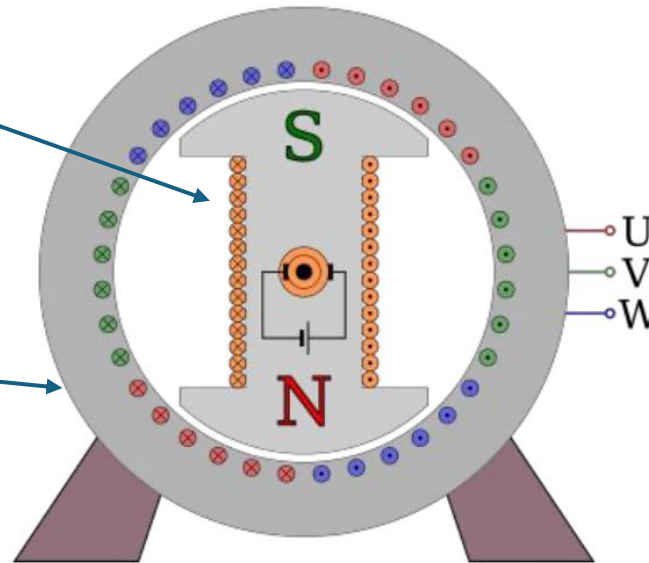
three alternating currents of the same frequency and amplitude but phase-shifted by 120° from each other

motivation:

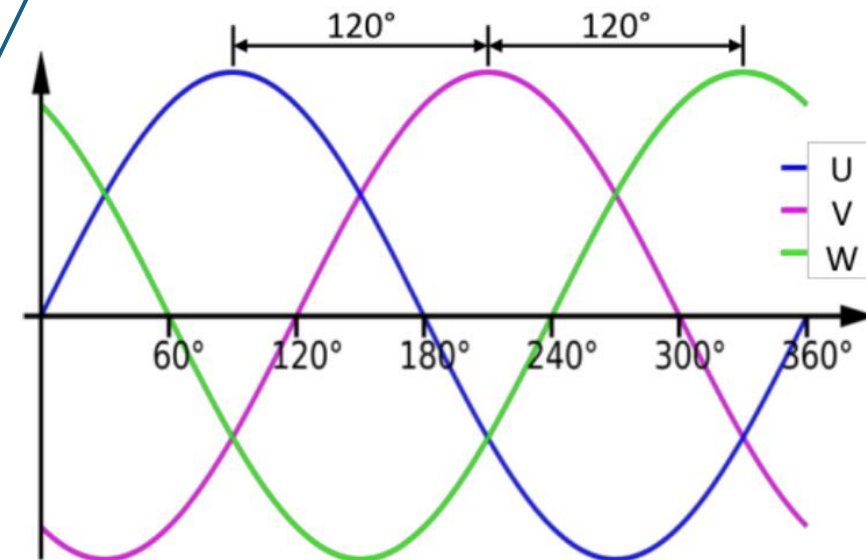
- constant power transmission
- generation of rotating magnetic field → electric motor

Three-Phase Generator

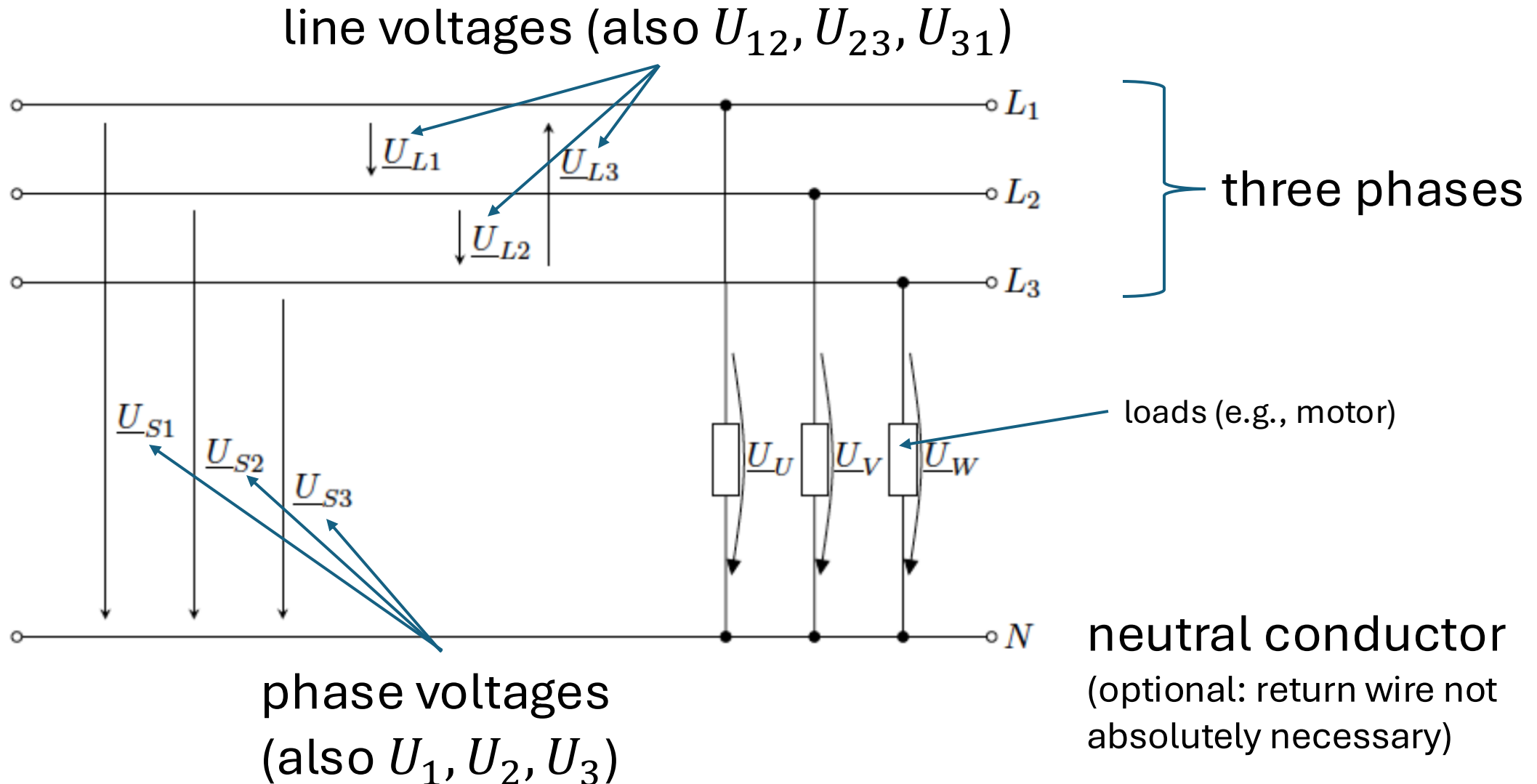
- rotor: (mechanically) rotating magnetic field (generated by coil with DC or permanent magnet)
- stator: three coils offset by 120°
- induces sinusoidal alternating voltage in each coil, phase-shifted by 120°



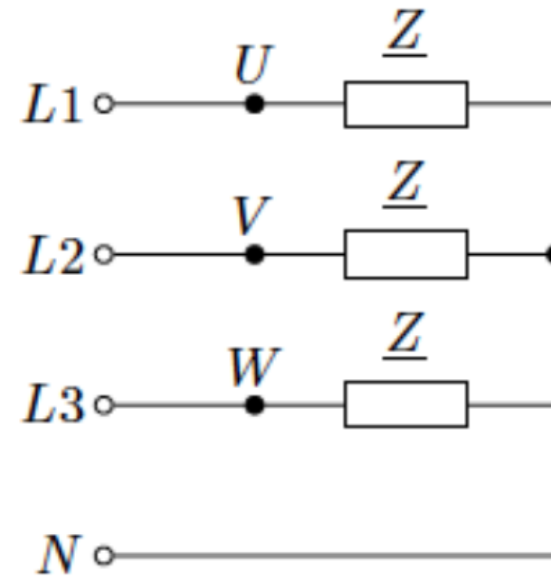
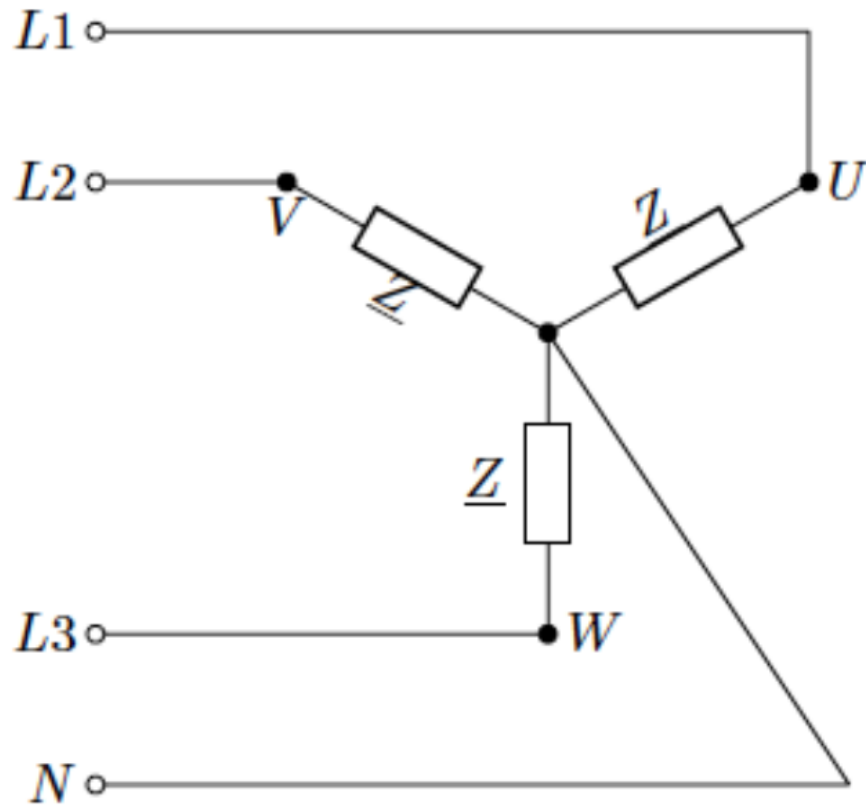
phase identifiers on motor terminals



Three-Phase Network



Star (Y) Connection



no current in N for symmetric loads (the three currents add to 0)
→ N could be removed

Phase Voltages

$$u_1 = \hat{u} \cdot \cos(\omega t)$$

$$u_2 = \hat{u} \cdot \cos\left(\omega t - \frac{2}{3}\pi\right)$$

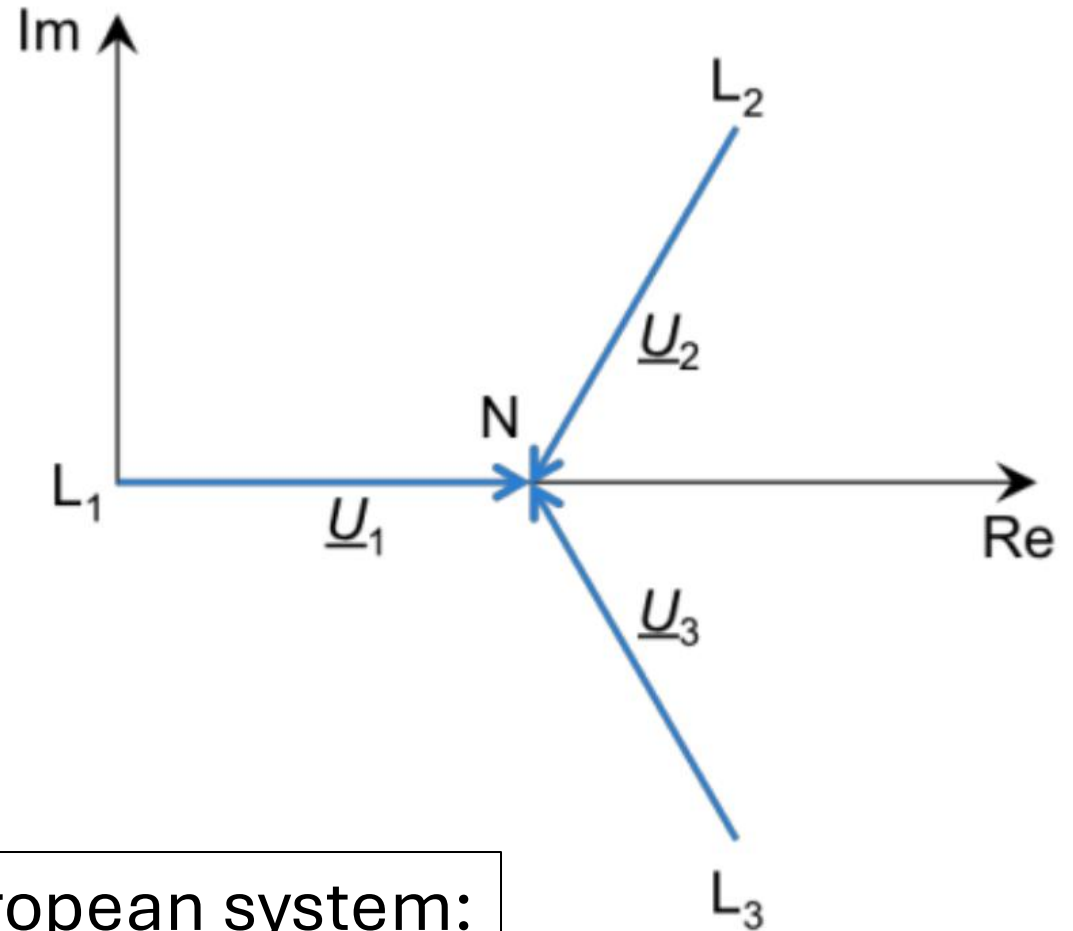
$$u_3 = \hat{u} \cdot \cos\left(\omega t - \frac{4}{3}\pi\right)$$

$$\underline{U}_1 = U$$

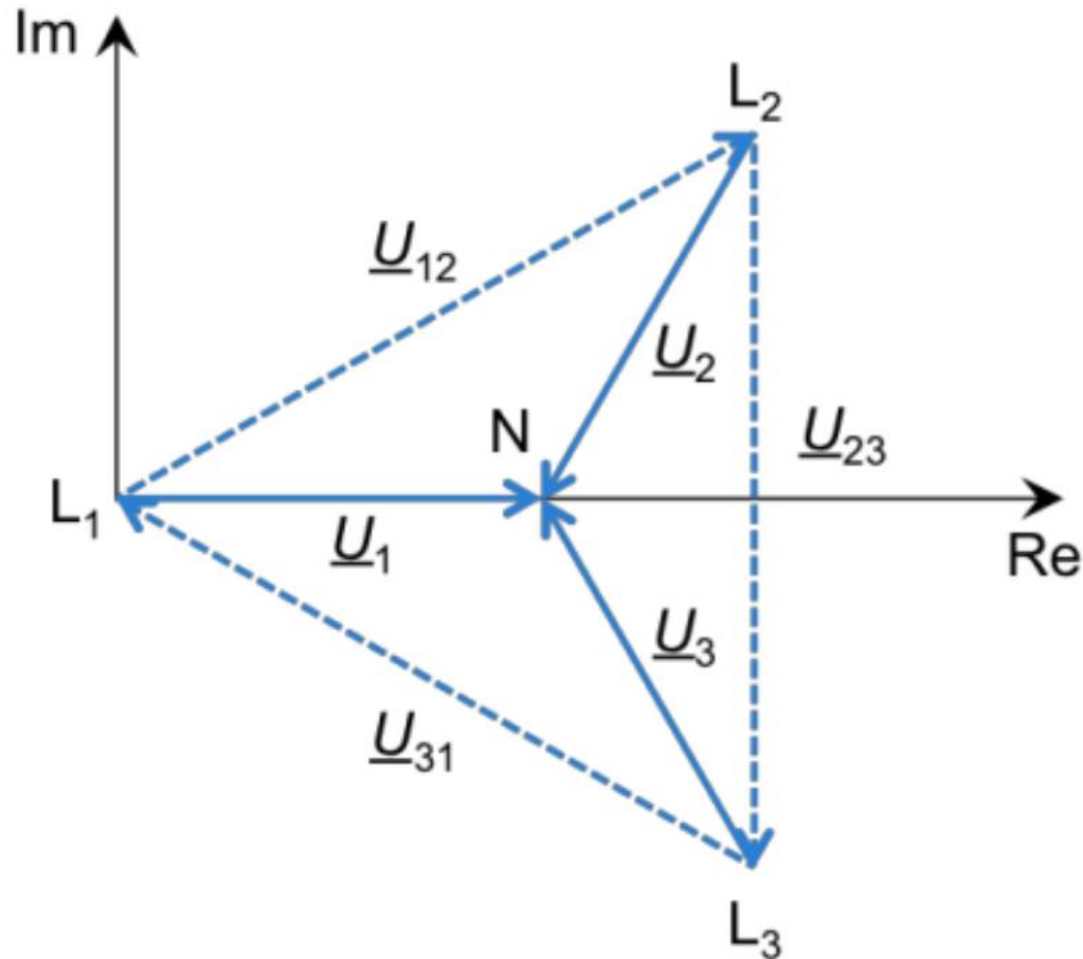
$$\underline{U}_2 = U \cdot e^{-j\frac{2}{3}\pi}$$

$$\underline{U}_3 = U \cdot e^{-j\frac{4}{3}\pi}$$

typical European system:
230 V



Line Voltages

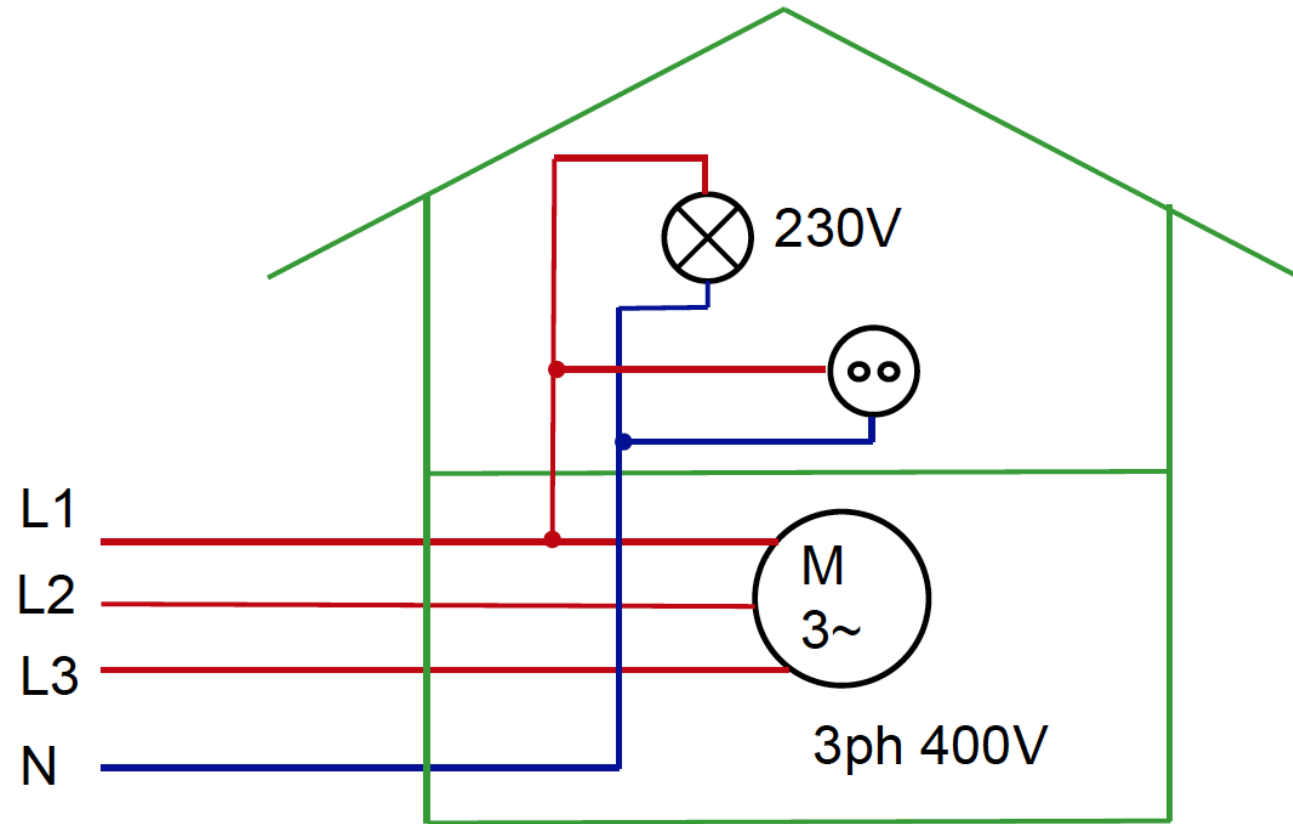


typical European system:
400 V

$$U_{12} = U_{23} = U_{31} = U_L = \sqrt{3} \cdot U$$

for example:

$$\begin{aligned}\underline{U}_{12} &= \underline{U}_1 - \underline{U}_2 = U - U \cdot e^{-j\frac{2}{3}\pi} = U \cdot \left(1 - \left(-0.5 - j\frac{\sqrt{3}}{2} \right) \right) \\ &= U \cdot 1.5 + j\frac{\sqrt{3}}{2}U = \sqrt{3} \cdot U \cdot e^{j\frac{\pi}{6}}\end{aligned}$$



Power for Symmetrical Load

$$\underline{I_1} = \underline{U_1} / \underline{Z}$$

$$\underline{I_2} = \underline{U_2} / \underline{Z}$$

$$\underline{I_3} = \underline{U_3} / \underline{Z}$$

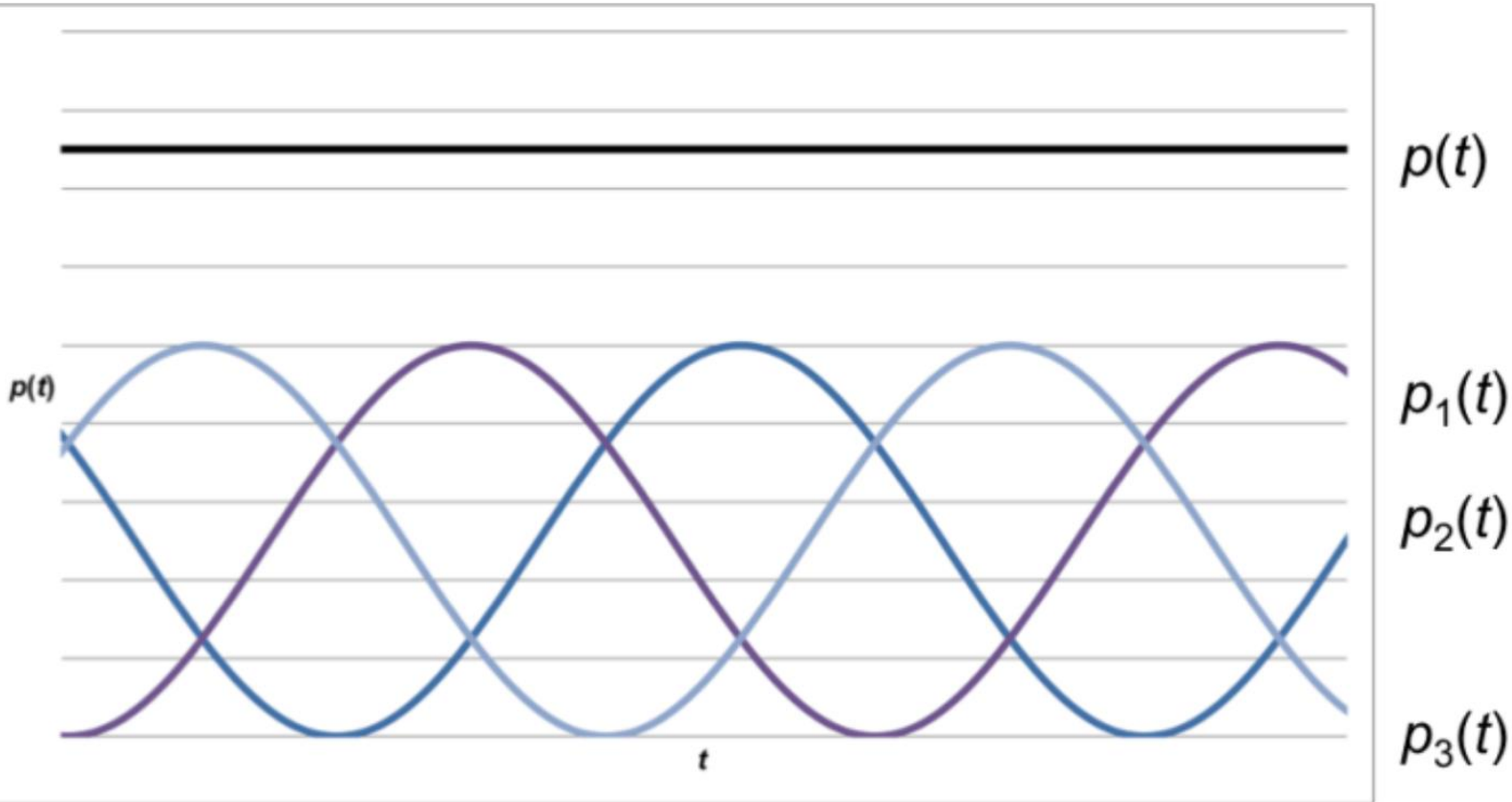
$$I_1 = I_2 = I_3 = I = I_L$$

$$P = 3 \cdot U_1 \cdot I_1 \cdot \operatorname{Re}(e^{j(\varphi_u - \varphi_i)}) = 3 \cdot U \cdot I \cdot \cos \varphi = \sqrt{3} \cdot U_L \cdot I_L \cdot \cos \varphi$$

$$Q = 3 \cdot U_1 \cdot I_1 \cdot \operatorname{Im}(e^{j(\varphi_u - \varphi_i)}) = \sqrt{3} \cdot U_L \cdot I_L \cdot \sin \varphi$$

$$S = \sqrt{P^2 + Q^2} = \sqrt{3} \cdot U_L \cdot I_L$$

Constant Power



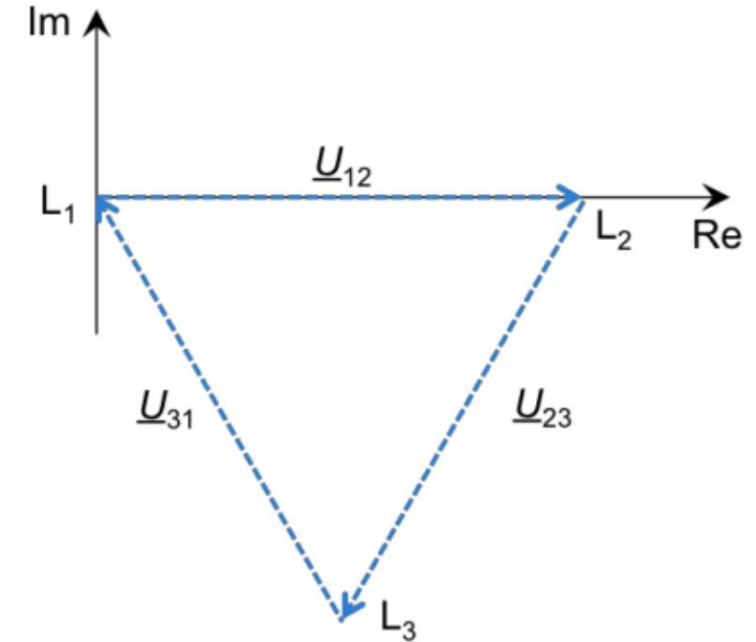
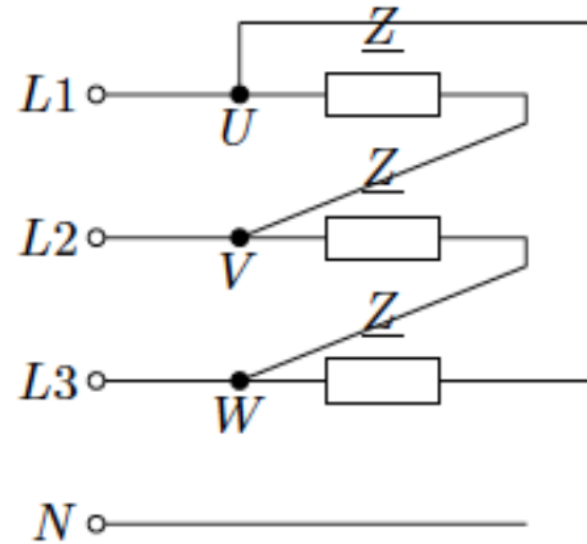
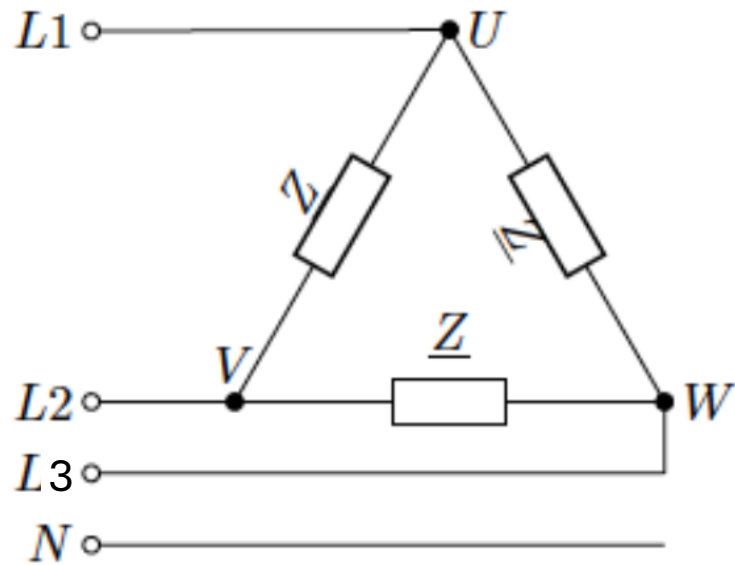
$$p(t) = \frac{u_1^2(t)}{R_1} + \frac{u_2^2(t)}{R_2} + \frac{u_3^2(t)}{R_3}$$

for symmetrical, purely resistive load:

$$R_1 = R_2 = R_3 = R$$

$$\rightarrow p(t) = \frac{\hat{U}^2}{R} \left(\cos^2 \omega t + \cos^2 \left(\omega t - \frac{2}{3} \pi \right) + \cos^2 \left(\omega t + \frac{2}{3} \pi \right) \right) = \frac{3\hat{U}^2}{2R} = \frac{3U^2}{R}$$

Delta (Δ) Connection



no neutral point

$$U_{12} = U_{23} = U_{31} = U_L = U = 400 \text{ V}$$

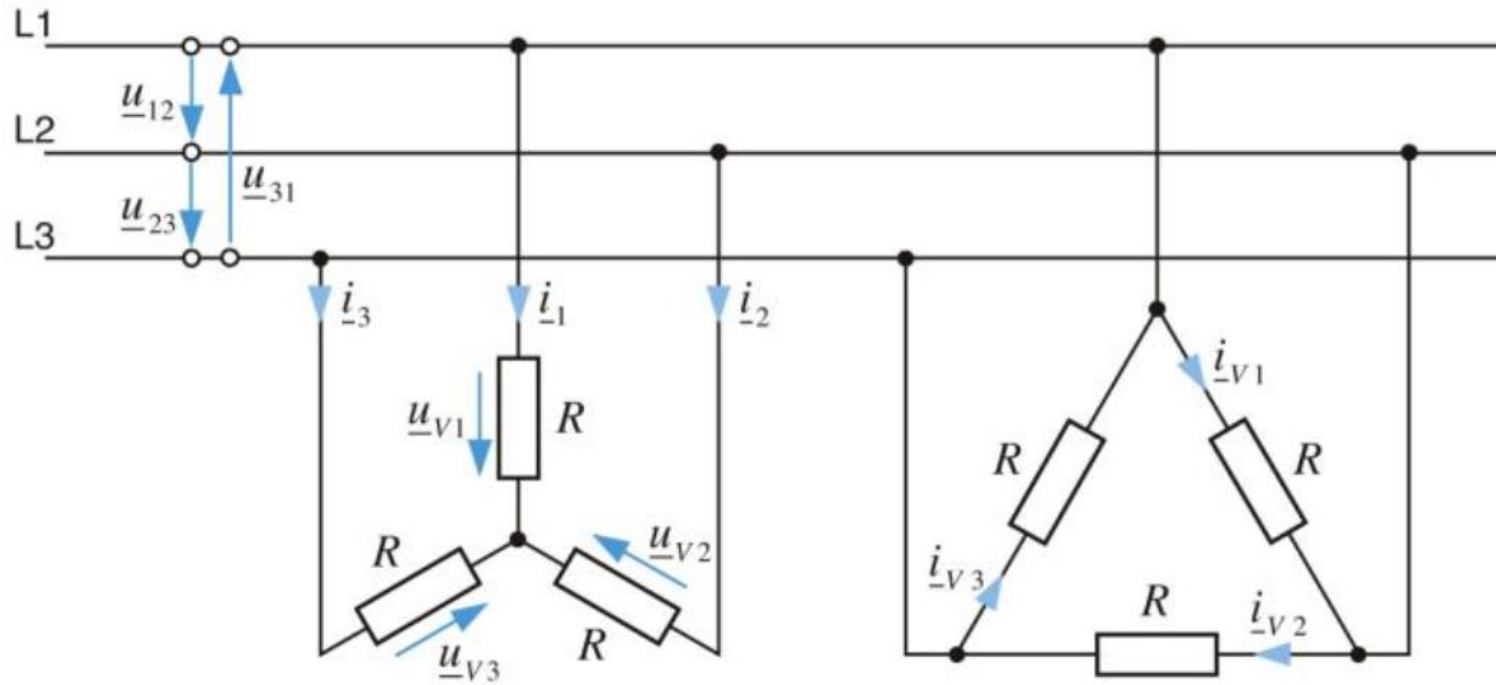
$$\underline{I}_{12} = \underline{U}_{12} / \underline{Z}$$

$$\underline{I}_{23} = \underline{U}_{23} / \underline{Z}$$

$$\underline{I}_{31} = \underline{U}_{31} / \underline{Z}$$

$$I_L = \sqrt{3} \cdot I$$

Power Consumption



for both star and delta:

$$P = \sqrt{3} \cdot U_L \cdot I_L \cdot \cos \varphi = 3 \cdot U \cdot I \cdot \cos \varphi$$

but for star:

$$U = U_L / \sqrt{3}, \quad I = U / R$$

$$\rightarrow P = \frac{U_L^2}{R} \cdot \cos \varphi$$

while for delta:

$$U = U_L, \quad I = U_L / R$$

$$\rightarrow P = 3 \cdot \frac{U_L^2}{R} \cdot \cos \varphi$$

$$\rightarrow P_{\Delta} = 3 \cdot P_Y$$